

Crosslight Software

Tutorial Workbook

for

LASTIP, APSYS, and PICS3D

Release 2003.04

by

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1. INTRODUCTION

This workbook helps new users to get familiar with the Crosslight Software products LASTIP, APSYS, and PICS3D. Simple exercises are described in detail and the reader is guided step by step through the simulation process. All explanations refer to the software release in March 2003 (2003.03) including the new graphical user interface SimuLASTIP, SimuAPSYS, and SimuPICS3D, respectively, for Windows 95/98/NT/ME/2000/XP. Each of the following sections describes a separate exercise which may take several hours to complete.

Like most other advanced device simulation software, Crosslight Software uses a variety of files for data input and output. These files are the most direct way of communicating with the software. All input files can be edited by the user. The graphical user interface makes it easy to deal with these files. All the files of a device simulation have the same file name (chosen by the user) but different file extensions. We will first review these file types and their function (Fig. 1). Software statements and command options are described in the reference manuals which should be opened in a separate window of the computer screen.

To perform a device simulation, the software simply needs to know how the device looks like and what kind of device characteristics it should calculate. This information is provided by two main input files, name.layer and name.sol. The .layer file contains the device structure, layer by layer, given in growth direction, including material compositions, doping, and geometry of the 2D cross section. By processing the .layer file, the software first generates sub-files which separate the information on geometry (.geo), doping (.doping), and materials (.mater) for later use. From the .geo file, the software then generates a file containing the 2D mesh (.msh). PICS3D requires another input file (.gain) which gives details on the calculation of the optical gain. Finally, the .sol file contains additional information needed for the simulation of the optical waveguide and the heat flux. The .sol file also lists all the commands and parameters of the actual simulation. The three input files .layer, .gain, and .sol contain all the information needed. They must be generated by the user. All other input files are generated by the software.

Output files can be separated in two groups: (1) files that contain the data generated by the software and (2) files listing commands to plot these data. The first group is quite large (Fig. 1) and it is discussed in the following chapters. The second group is quite small. The file name.mplt

is generated automatically from name.layer and it plots the mesh. The file name.gain is not only an input file, it can also be used to plot the optical gain spectrum – even before the actual device simulation is performed. The main plot file is name.plt which needs to be generated by the user. It contains all the commands to visualize the final results of the simulation.

<i>main files generated by the user</i>	<i>secondary files generated by the software</i>	<i>function,content</i>
name.layer →	name.mater name.doping name.geo name.mplt (name.vcsl) → name.msh	materials, properties doping profile geometry, numerical mesh mesh plotting optical sections of VCSEL
name.gain		computes and plots gain
name.sol →	(name.rtd) (name.stw) (name.big) name.out_# (name.zdata_#) (name.zlog_#) name.std_# name.log	results on optical cavity standing wave (VCSEL) transparency current xy-result data z-result data z-result log CrosslightView input file runtime output
name.plt		plots results

Fig. 1 Overview of the software file types. The file names in parentheses only apply to PICS3D.

2. HOW TO RUN THE SIMULATION EXAMPLES PROVIDED (LASTIP)

The examples directory contains many applications which offer an easy way to evaluate the capabilities of the software. These examples also offer a convenient starting point to develop new device simulations. Running one of these example simulations is very easy. Here, we will simulate an simplified quantum well laser diode using LASTIP (Fig.2). The input files of this example are located in the directory `c:\crosslig\lastip_examples\A_layered_LD`.

- (1) Activate SimuLastip. You can do this by simply choosing "SimuLastip" located in the LASTIP program folder (`c:\crosslig\lastip`). Once SimuLastip is open, choose "File" from the menu bar and scroll down to "Open Project". Go to the folder `C:\crosslig\lastip_examples\A_layered_LD`. You should see the files `gaas10.gain` and `gaas10.sol`. Select `gaas10.sol` and open it. You should see something like Fig. 3. It gives the main simulation file in the right-hand-side window and a list of all other files in the left-hand-side window. Double-click on `gaas10.layer` file to open it. The file contains the layer structure of this laser. Click on any of the blue commands and go to `Edit > Wizard (Ctrl-W)`. This will bring up the command editor which explains the command options in detail. As an advanced user, you could edit the input files now. For beginners, we skip this point. Click Cancel.

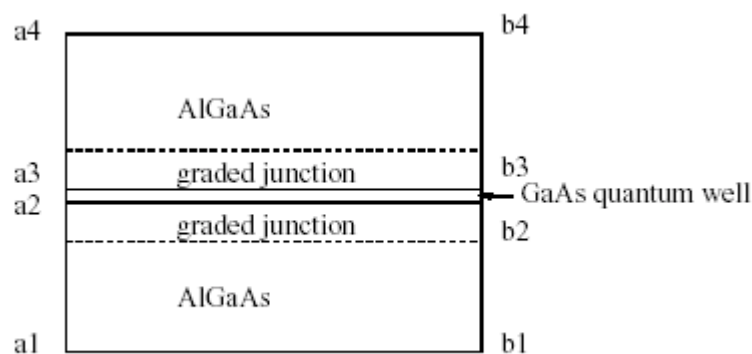


Fig. 2 Tutorial example laser (gaas10).

- (2) Using the `.layer` file, we first generate new files containing information on device geometry (`.geo`), doping (`.doping`), materials (`.mater`), as well as the file `.mplt` to plot the mesh. We will simply overwrite the files that already exist in this example directory. Click on "Layer File

(gaas10)” using the right mouse button and select “Process and generate .geo”. The bottom window shows the processing status.

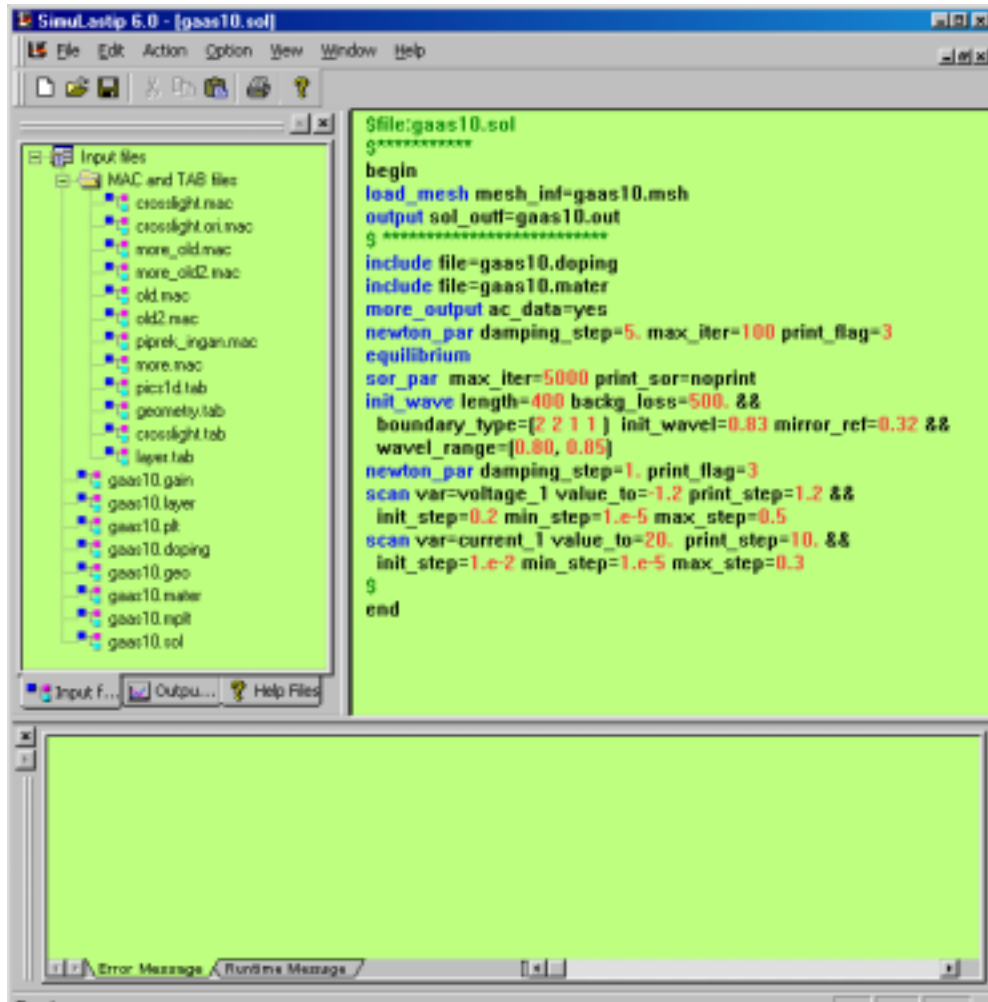


Fig. 3 SimuLastip user interface.

- (3) From the device geometry file (.geo), we now generate the mesh file (.msh) later needed by the main simulator. Right-click on “gaas10.geo” and select “Generate Mesh”. To view the mesh, right-click on “gaas10.mplt” and select “View Mesh (mplt)”. If GhostView is installed on your computer (available at www.crosslight.com), you will see a simple zigzag mesh for our 1D example structure (Fig. 4). Close the window. The plot will be saved as gaas10_m.ps and you can display it again by selecting it from the “Output” file listing. Alternative plot options can be selected inside gaas10.mplt (right window), by replacing “plot_device=postscript” with “plot_device=window”, for example, in order to use the built-in display.

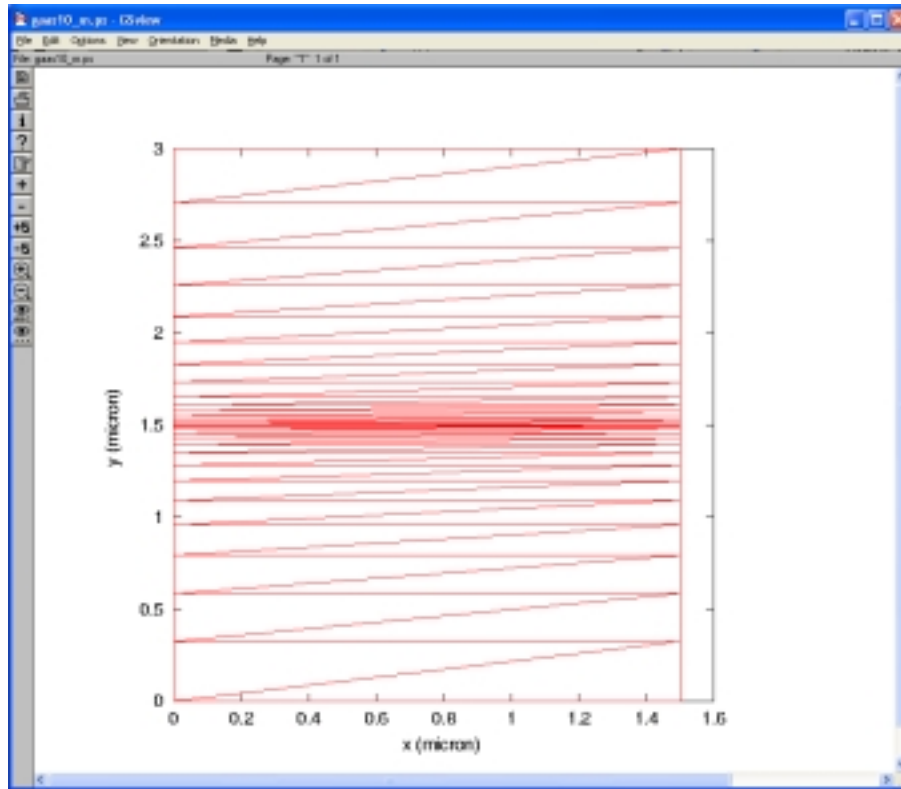


Fig. 4 Mesh plot.

- (4) It is always helpful to check out the optical gain produced by the laser's active region before starting the full simulation. This can be done using the `gaas10.gain` file which already exists in our case. You can process it by right-clicking on "`gaas10.gain`" and selecting "Process Gain File". You will first see gain spectra for different carrier densities (Fig. 5a). It is always important to have positive gain at the intended laser emission wavelength (here $0.83\mu\text{m}$). Clicking on "+" in the GSview side panel shows the peak gain vs. carrier density (Fig. 5b). This figure reveals a transparency carrier density of about $1.7 \times 10^{18} \text{cm}^{-3}$ for our laser. Close the GSview window, both figures are saved as `gaas10_g.ps`

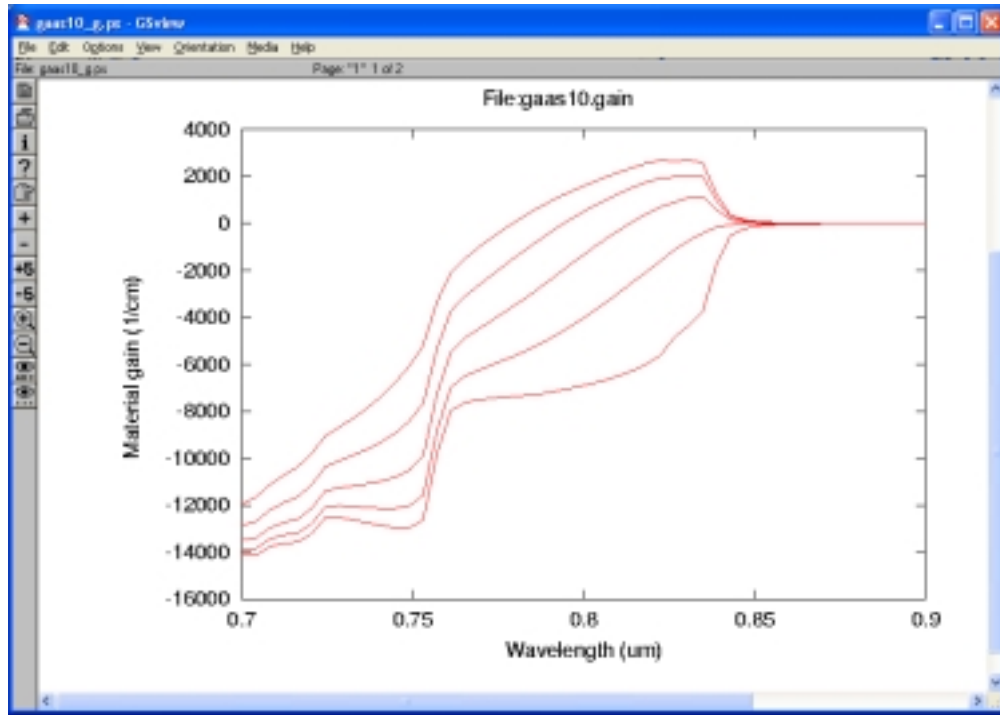


Fig. 5a Gain spectra plotted at different carrier densities.

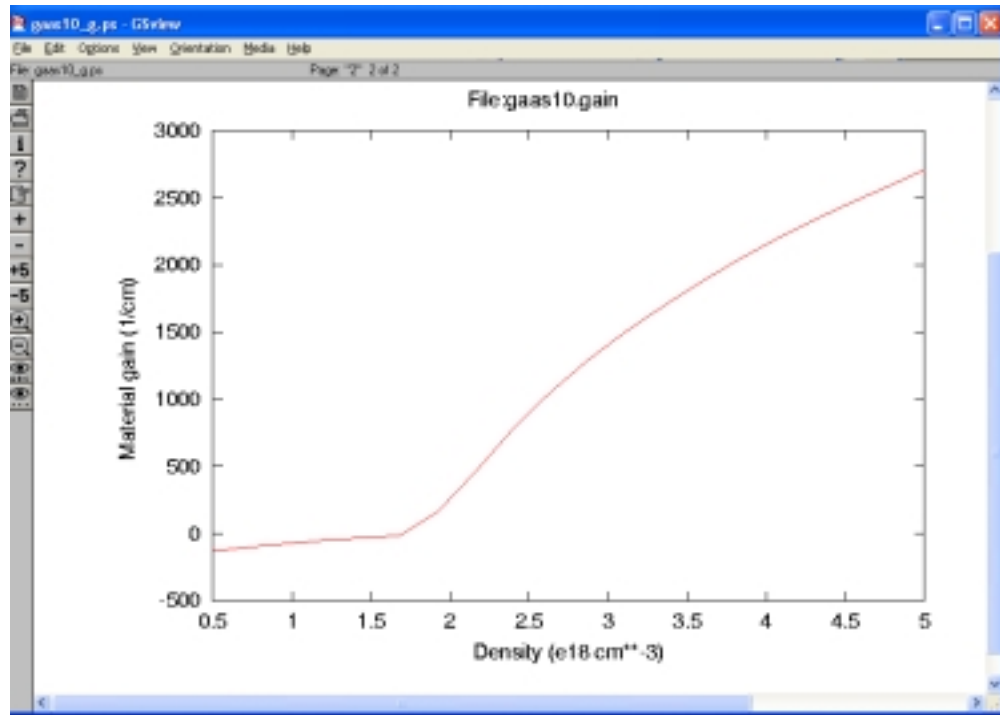


Fig. 5b Peak gain vs. carrier density.

- (5) You can now start the main simulation using the `gaas10.sol` file. Right-click on “`gaas10.sol`” and select “Start Simulation”. The simulation ramps up the injection current according to the bias points given in `gaas10.sol`. It can take from a few minutes to a couple of hours depending on the complexity of the device and on the number bias points. Using a 2GHz Pentium PC, our example runs for about 2 minutes. The progress window displays the runtime output which will eventually be saved as `gaas10.log` (click on “Save” in the pop-up window). After the simulation has finished, you see several output files in the left “Output” list of SimuLastip. The `.std_###` files are used by the interactive plotting software CrosslightView which is not covered in this tutorial (see GUI reference manual). We here rather use the files `gaasa10.out_###` which contain the same information and which are accessed via the `gaas10.plt` file.
- (6) To plot your results, right-click on “`gaas10.plt`” and select “View Results”. You will see the following plots (click on “+” or “-” to switch): energy band diagram (Fig. 6a), wave intensity (Fig. 6b), current-voltage (IV) curve (Fig. 6c), light-current (LI) curve (Fig. 6d), as well as the intrinsic AC capacitance vs. frequency characteristic (Fig. 6e). In the next section, you will learn how to edit the plot file to include other plots.

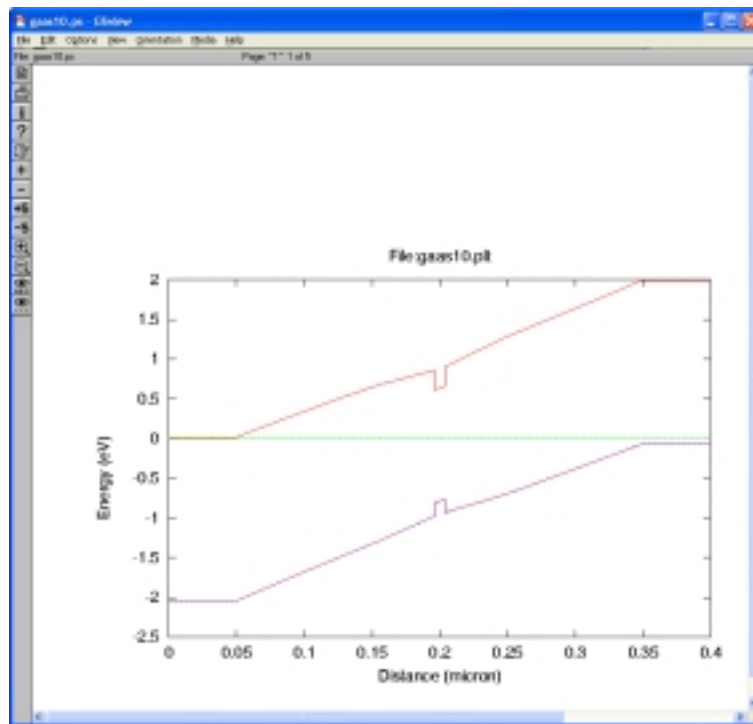


Fig. 6a: Vertical band diagram at zero bias (red: conduction band , blue : valence band)

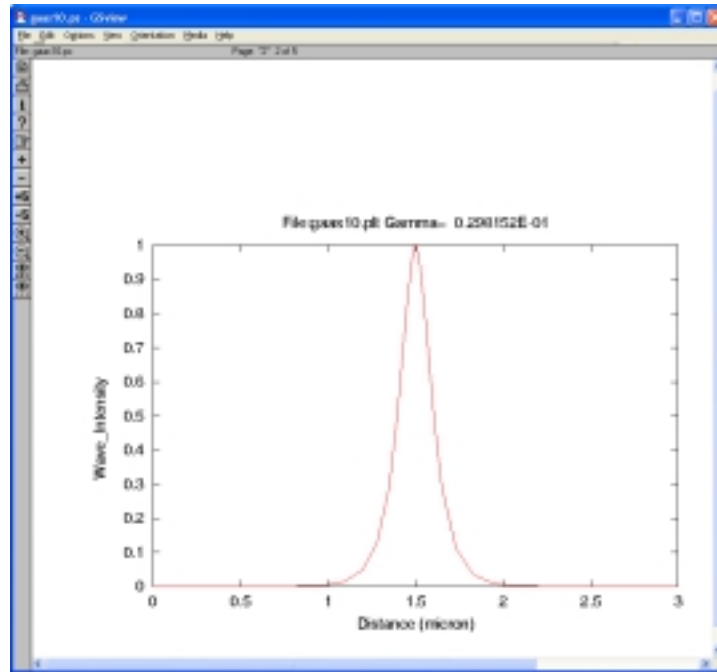


Fig. 6b: Vertical profile of the optical wave. Also given is the optical confinement factor $\Gamma=0.0298$ for our quantum well structure.

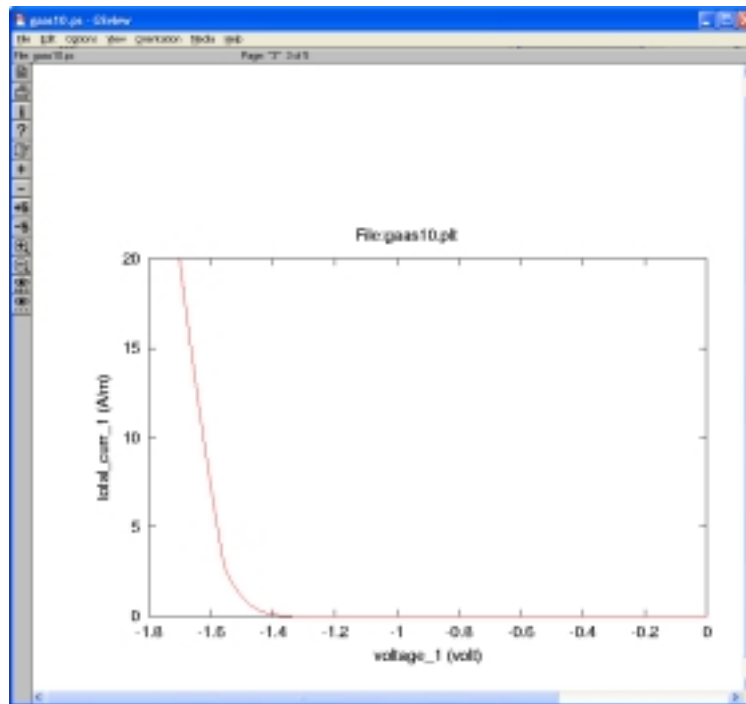


Fig. 6c: Current vs. voltage characteristic of example gaas10.

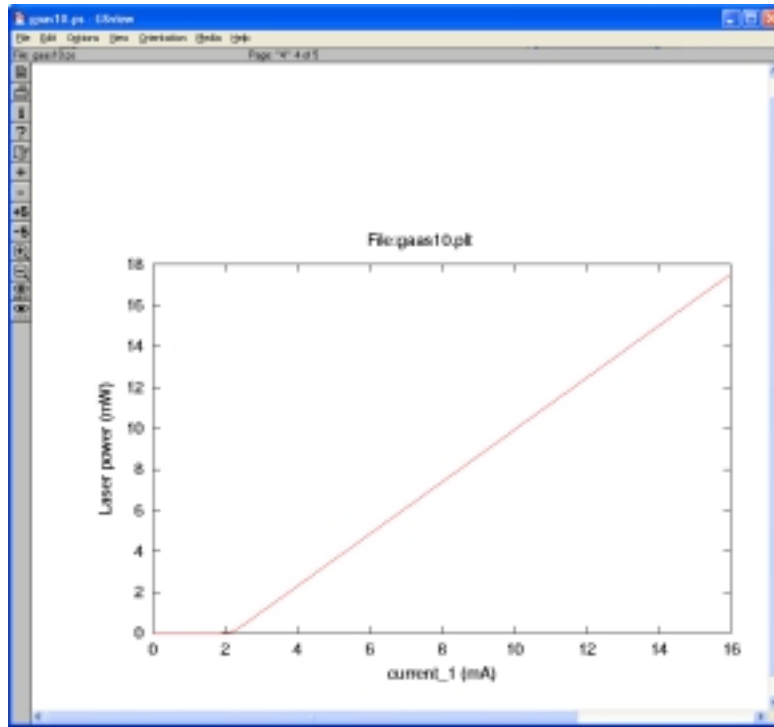


Fig. 6d: Laser power (right) vs. current. The threshold current is about 2mA.

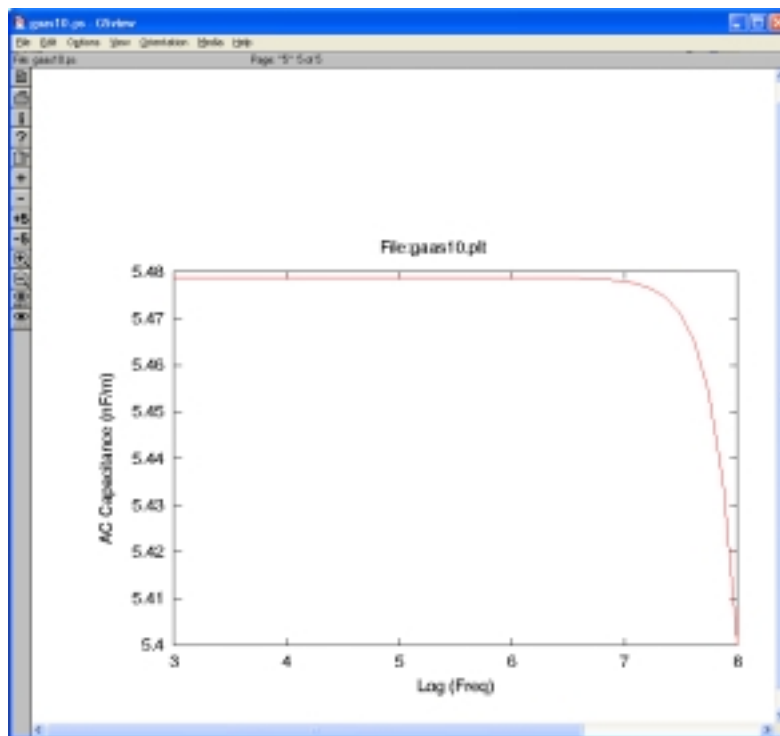


Fig. 6e: Intrinsic AC capacitance vs. frequency.

3. HOW TO GENERATE INPUT FILES FOR AN IN-PLANE LASER (LASTIP)

Our tutorial example is a Graded-Index Separate-Confinement Heterostructure (GRIN-SCH) laser diode, emitting at $0.83\mu\text{m}$ wavelength. It is the same device as in the previous section (Fig. 2). For simplicity, we restrict this example to the center part of the laser and neglect any lateral non-uniformity, i.e., only the active stripe is considered ($3\mu\text{m}$ wide, $400\mu\text{m}$ long). Since the laser is symmetric, the simulation can be restricted to the right half (width $1.5\mu\text{m}$). The vertical layer sequence is given in the following table:

Thickness [micron]	Doping [m^{-3}]	Material	Comment
1.35	1×10^{24} p	$\text{Al}_{0.71}\text{Ga}_{0.29}\text{As}$	p-doped upper cladding
0.1462	none	$\text{Al}_x\text{Ga}_{1-x}\text{As}$	Grading layer ($x=0.33\dots0.71$)
0.0076	none	GaAs	GaAs quantum well active layer
0.1462	none	$\text{Al}_x\text{Ga}_{1-x}\text{As}$	Grading layer ($x=0.71\dots0.33$)
1.35	1×10^{24} n	$\text{Al}_{0.71}\text{Ga}_{0.29}\text{As}$	n-doped lower cladding

Tab. 1 Layer structure of example gaas10.

- (1) Start "SimuLastip" located in the LASTIP programs folder. From the menu, select "File > New Project". Choose any name other than gaas10, e.g., "mylaser" and the directory "C:\crosslig\lastip_examples\mylaser". Select only the .layer file. The new file mylaser.layer will appear in the right-hand-side window, but it is still empty. Right-click on mylaser.layer and select "EdgeSetup layer" (overwrite). An input dialogue will start requesting all data required for the simulation (Fig. 7a). The input goes layer by layer starting at the bottom. Take your time and answer each question carefully. A wrong answer may later cause trouble with the simulation. For your reference, all questions and answers are given in Appendix A1. However, try not to look up the answers and use the device information given above. For the active layer, the software requires two material macros containing different information (active: AlGaAs/AlGaAs and bulk: GaAs). At the end, pick 20 mesh points. The maximum average mesh spacing should be about 20% of the largest layer thickness, pick 0.2 in our example.

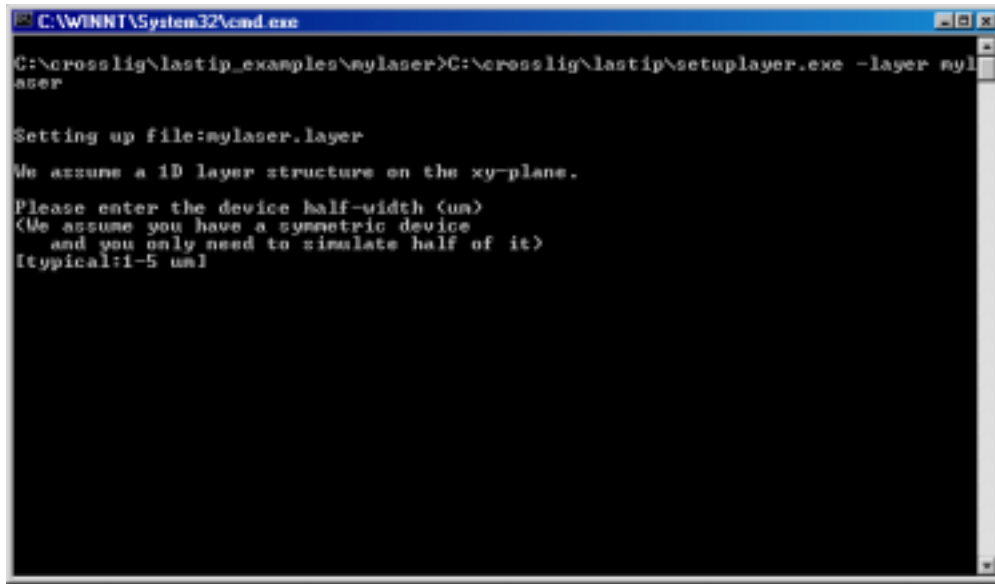


Fig. 7a: EdgeSetupLayer: MS-DOS dialog window.

```
$file mylaser.layer
begin_layer
column column_num=1 w= 0.150000E+01 mesh_num=2 r=1.
bottom_contact column_num=1 from=0 to= 0.150000E+01 &&
  contact_num=1 contact_type=ohmic
$
layer_mater macro_name=algaas      &&
  var1= 0.710000E+00 &&
  column_num=1
layer d= 0.135000E+01 n= 16 &&
  n_doping1= 0.1000E+25 &&
  r= 0.8000E+00 xp1=0 xp2=1
$
layer_mater macro_name=algaas      &&
  var1= 0.000000E+00 &&
  grade_var=1 &&
  grade_from= 0.710000E+00 grade_to= 0.330000E+00 &&
  column_num=1
layer d= 0.146200E+00 n= 6 &&
  shift_center= -0.4873E-01 &&
  r= -0.1200E+01 xp1=1 xp2=1
```

```

$
layer_mater macro_name=gaas      &&
  active_macro=AlGaAs/AlGaAs    &&
  avar1= 0.000000E+00 &&
  avar2= 0.330000E+00 &&
  avar3= 0.300000E+03 &&
  column_num=1
layer d= 0.760000E-02 n= 6 &&
  r= -0.1200E+01 xp1=1 xp2=1
$
layer_mater macro_name=algaas    &&
  var1= 0.000000E+00 &&
  grade_var=1 &&
  grade_from= 0.330000E+00 grade_to= 0.710000E+00 &&
  column_num=1
layer d= 0.146200E+00 n= 6 &&
  shift_center= 0.4873E-01 &&
  r= -0.1200E+01 xp1=1 xp2=1
$
layer_mater macro_name=algaas    &&
  var1= 0.710000E+00 &&
  column_num=1
layer d= 0.135000E+01 n= 16 &&
  p_doping1= 0.1000E+25 &&
  r= 0.1200E+01 xp1=1 xp2=0
$
top_contact column_num=1 from=0 to= 0.150000E+01 &&
  contact_num=2 contact_type=ohmic
end_layer

```

Fig. 7b: Resulting file mylaser.layer.

- (2) As result of this dialogue, you should see the file mylaser.layer as shown in Fig. 7b. To process this file, right-click on it and select “Process and generate .geo”. This generates new files containing information on device geometry (.geo), doping (.doping), materials (.mater), as well as the file .mplt to plot the mesh. To create the mesh file, right-click on “mylaser.geo” and select "Generate Mesh" (in case of an error message, close the project and open it again).

Plot the mesh by right-clicking on “mylaser.mplt” and selecting “View Mesh”. Compared to Fig. 4, your mesh probably has much less grid points. Close the GSview window.

- (3) It is usually required to adjust the mesh by editing the .layer file. The number of mesh points per layer is given by the option “n=” in the “layer” command line. For each of your five layers, double this number in your file mylaser.layer. Save the file, process it again and view the resulting mesh as described in (2). The mesh should now have a smaller grid spacing.
- (4) To check out the quantum well gain and photoluminescence spectra, the mylaser.gain file needs to be generated first. Right-click on “mylaser.gain”, select “Setup by command” (in case of a “sharing violation error”, close the existing empty file in the right window). A MS-DOS dialog window appears asking for wavelength, carrier density, temperature and plot options (see Appendix A2). Select the plot options 1,2, and 6. The final gain file is shown in Fig. 8a.

```
$file:mylaser.gain
begin_gain
plot_data plot_device=postsript
temperature temp= 0.3000E+03
include file=mylaser.mater
gain_wavel wavel_range=[ 0.8100E+00 0.8500E+00] &&
  conc_range=[ 0.5000E+24 0.1000E+26] &&
  curve_number=20
sp.rate_wavel wavel_range=[ 0.8100E+00 0.8500E+00] &&
  conc_range=[ 0.5000E+24 0.1000E+26] &&
  curve_number=20
gain_density conc_range=[ 0.5000E+24 0.1000E+26] &&
  wavel_range=[ 0.8100E+00 0.8500E+00] &&
  data_point=30
end_gain
```

Fig. 8a: Gain file mylaser.gain

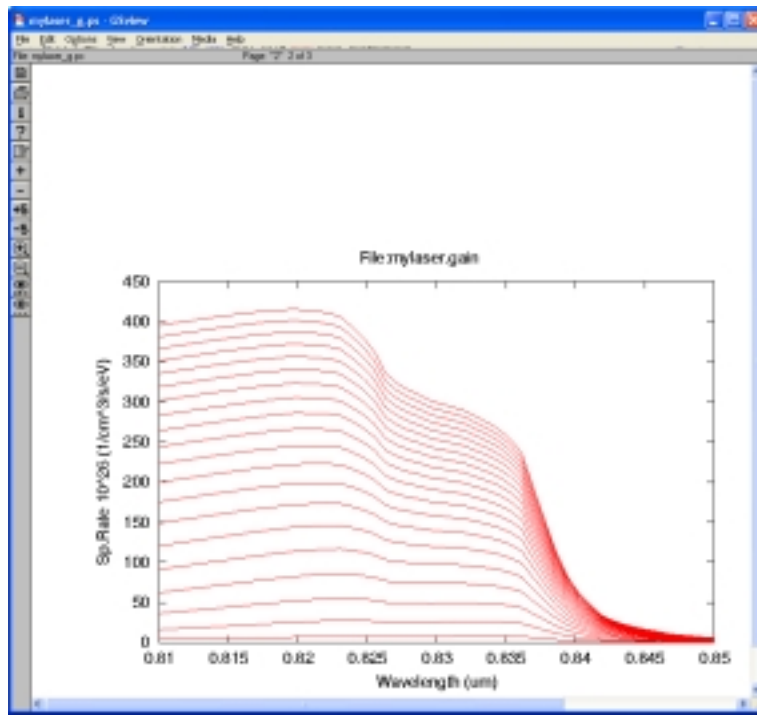


Fig. 8b: Spontaneous emission spectra at different carrier densities.

- (5) To process the gain file, right-click on it and select “Process Gain File”. The first plot is similar to Fig. 5a with a narrower wavelength window and more carrier density values. The second plot shows the spontaneous emission spectrum at the same carrier densities (Fig. 8b). As expected, it peaks at slightly shorter wavelength. The third plot shows gain vs. carrier density similar to Fig. 5b. After closing the GSview window, you may double-click on the gain file to change the parameter range of these plots.

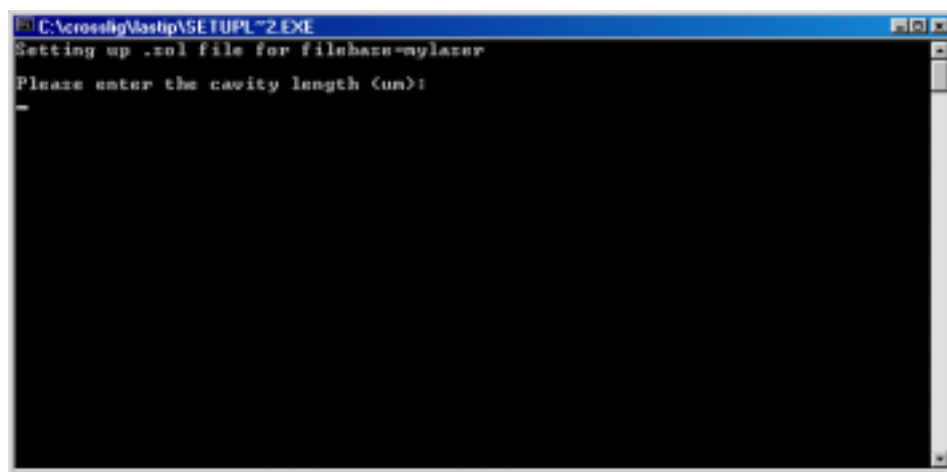


Fig. 9a: MS-DOS window for setup of the .sol file.

(6) The next step is to generate the final input file `mylaser.sol`. Right-click on “mylaser.sol” and select “Setup by Command” (overwrite, OK on error message). You should see a window similar to Fig 9a. Use the information given at the beginning of this section. Select the 1D optical wave boundary. The current should be well above the threshold current, let's say 10mA for the full device (cf. Fig. 6d). The full dialogue is given in appendix A3 for your reference. After you are finished, double-click on “mylaser.sol” in order to see it (Fig. 9b). The “init_wave” command includes an option “backg_loss” that sets the modal optical loss, the default value is 500/m. The “mirror_ref” option sets the mirror reflectance (default 0.32). After clicking on “init_wave” you can select “wizard” from the “edit” menu to see and set additional options for this command (Fig. 9c). This is a powerful feature, however, the most up-to-date list of command options is given in the file `C:\crosslig\crosslight.tab`. Close the wizard for now.

```
$file:mylaser.sol
begin
load_mesh mesh_inf=mylaser.msh
include file=mylaser.mater
include file=mylaser.doping
output sol_outf=mylaser.out
newton_par damping_step=5. max_iter=100 print_flag=3
sor_par max_iter=3000 print_sor=noprint
init_wave point_ll=[ 0.0000E+00 0.0000E+00] &&
  point_ur=[ 0.1500E+01 0.3000E+01] &&
  fld_center=[ 0.0000E+00 0.1500E+01] &&
  length= 0.4000E+03 backg_loss=500. &&
  boundary_type=[2 2 1 1] init_wavel= 0.8300E+00 mirror_ref=0.32 &&
  wavel_range=[ 0.8100E+00 0.8500E+00]
equilibrium
newton_par damping_step=1. print_flag=3
scan var=voltage_1 value_to= -0.1345E+01 print_step= 0.1345E+01 &&
  init_step= 0.2689E+00 min_step=1.e-5 max_step=0.5
scan var=current_1 value_to= 0.1250E+02 print_step= 0.1250E+02 &&
  init_step= 0.1250E+00 min_step=1.e-5 max_step= 0.1250E+01
end
```

Fig. 9b: Resulting file mylaser.sol

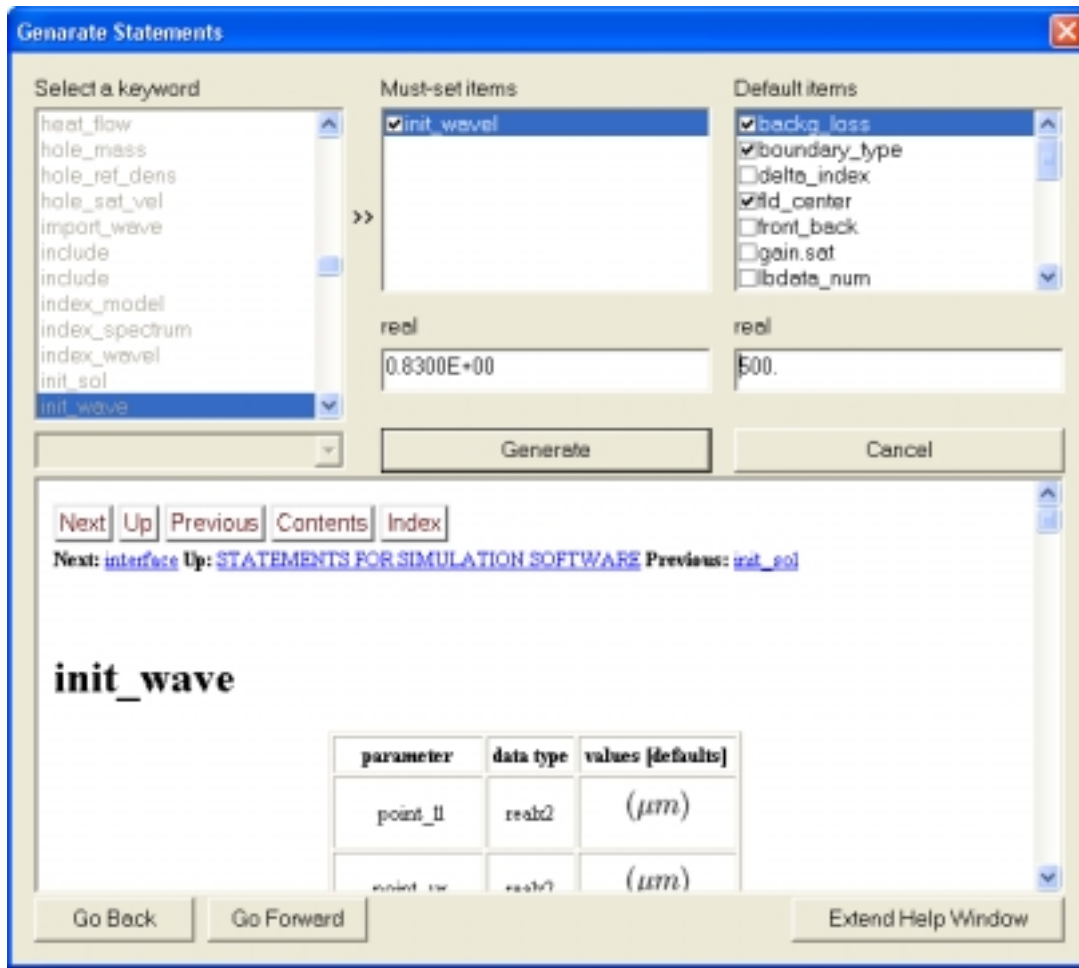


Fig. 9c: Wizard editor for software commands.

- (7) You can now start the actual simulation of your laser. Choose "Action > Main Simulation > Start the simulation". It should take only a few minutes. The run-time output is saved as mylaser.log. The simulation always ends by generating *.std files. If this does not happen, look up the mylaser.log file for error messages at the end. Compare your .layer and .sol files to the equivalent gaas10.* files and make changes if necessary. Click on "Output files" at the bottom of the left window and make sure that you have 3 files mylaser.out_*. These are the main output files investigated in the following.
- (8) To plot your results, you need to first generate the mylaser.plt file. To do so, right-click on "mylaser.plt" and select "Setup plt file". A MS-DOS window pops up again asking questions (in case of a "sharing violation error", close the existing empty file in the right window). You may select many different plot options. I suggest to start with the following vertical (y) plots at 1/2 of the device: energy bands (6) and local gain (29). You should also pick the L-I plot (see Appendix A4). After the MS-DOS window closes, double click on "mylaser.plt" to see the

file (Fig. 10). The commands are easy to understand, you may use “wizard” for further options. Note that “plot_scan” has default scaling factors of “scale_curr=2” and “scale_lit=2” which double current and power in the LI plot to account for the full device.

```
$file:mylaser.plt
begin_pstprc
plot_data plot_device=postsript
get_data main_input=mylaser.sol    &&
    sol_inf=mylaser.out    &&
    xy_data=[ 3 3] scan_data=[1 3]
plot_1d variable=band    &&
    from=[ 0.7500E+00 0.0000E+00] &&
    to=[ 0.7500E+00 0.3000E+01]
plot_1d variable=local_gain    &&
    from=[ 0.7500E+00 0.0000E+00] &&
    to=[ 0.7500E+00 0.3000E+01]
plot_scan scan_var=current_1 &&
    variable=laser_power
end_pstprc
```

Fig. 10 Plot file mylaser.plt

(9) Plot your curves by right-clicking on “mylaser.plt” and selecting “View results”. The band diagram is now shown at maximum bias, including quasi Fermi levels (Fig. 11). The following plot shows strong quantum well gain. The threshold current in the last plot should again be around 2mA. The LASTIP reference manual and its appendix gives all possible options for each plot command.

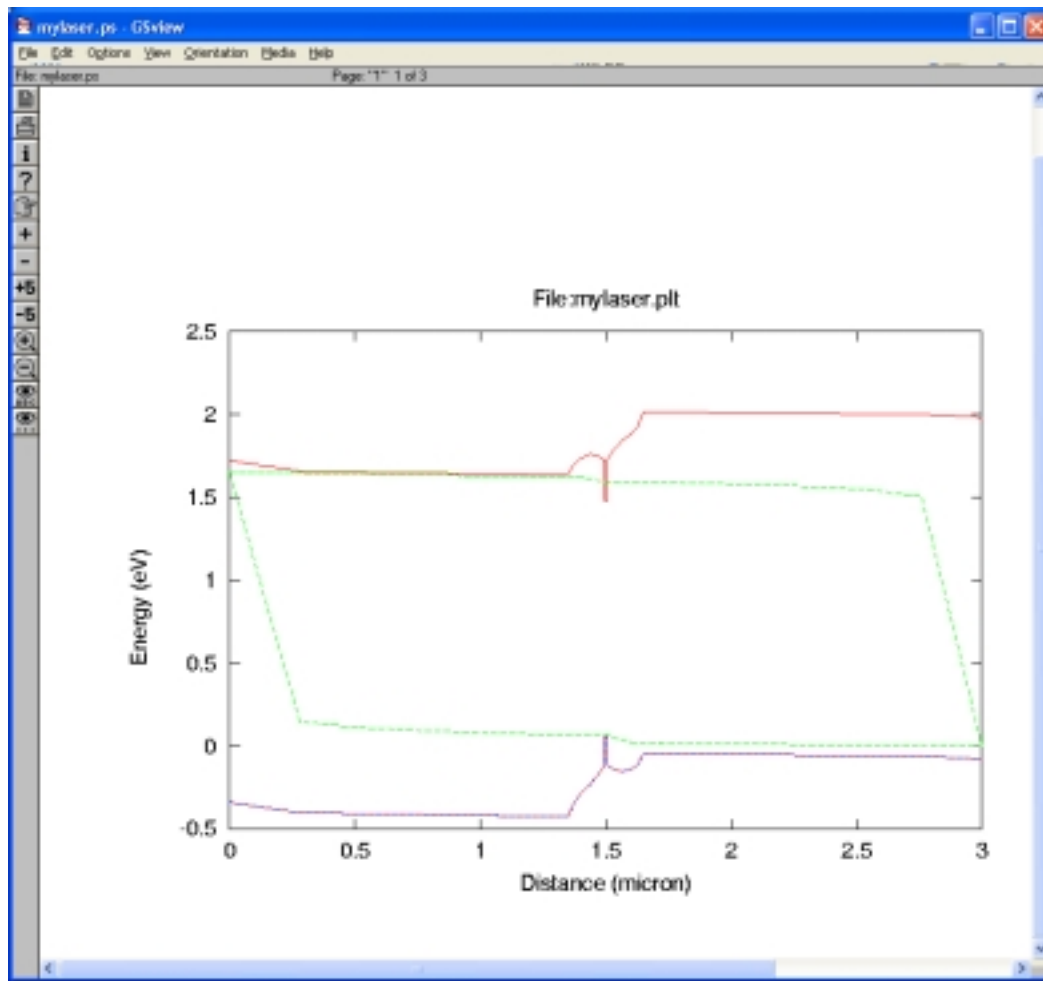


Fig. 11 Energy band diagram at maximum bias (green: quasi Fermi levels).

4. HOW TO SIMULATE A LIGHT EMITTING DIODE USING APSYS

The setup and the processing of a device simulation in APSYS works the same way as in LASTIP. We here use a GaN light emitting diode (LED) as example. The examples directory is C:\crosslig\apsys_examples\LED\gan_led. The LED is schematically shown in Fig. 12. This is now a 2D example as the current flows from the top contact to the side electrode.

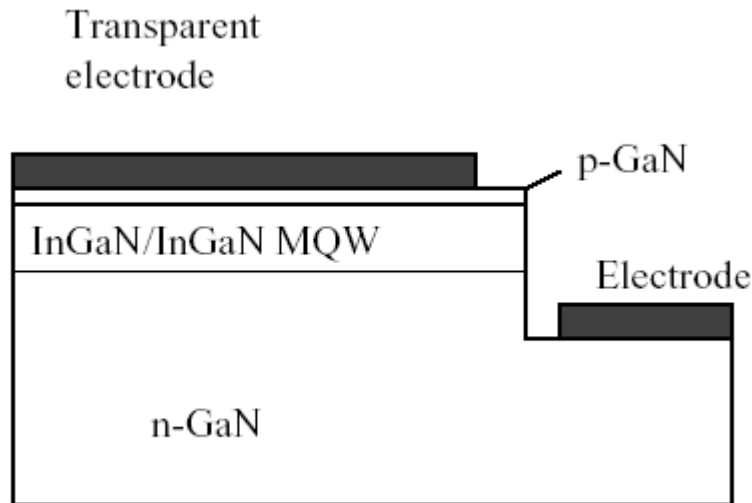


Fig. 12: GaN light emitting diode example ganled.

- (1) Activate SimuApsys. You can do this by simply choosing "SimuApsys" located in the APSYS program folder (c:\crosslig\apsys). Once SimuApsys is open, choose "File" from the menu bar and scroll down to "Open Project". Go to the example folder given above. Select ganled.sol and open the layer file (Fig. 13). The file gives two columns to describe our 2D diode. The left column 1 is 400 μ m wide and partially covered by the 390 μ m wide top contact. The right column 2 is 140 μ m wide and partially covered by the 120 μ m wide side contact. Both columns share only the bottom GaN layer, all other layers are only located in column 1 while the parallel layer in column 2 is void. The input for the 5 quantum wells is identical and given in the file ganled.qw. You may import this file instead of using "include file=ganled.qw". This reveals that the LED example assumes a 3nm thick $\text{In}_{0.4}\text{Ga}_{0.6}\text{N}$ quantum well with 5nm thick $\text{In}_{0.08}\text{Ga}_{0.92}\text{N}$ barriers. After clicking on any command, go to "Edit > Wizard" to evaluate further command options.
- (2) Using this .layer file, we first generate new files containing information on device geometry (.geo), doping (.doping), materials (.mater), as well as the file .mplt to plot the mesh. We will

simply overwrite the files that already exist in this example directory. Click on “ganled.layer” using the right mouse button and select “Process and generate .geo”. The bottom window shows the processing status.

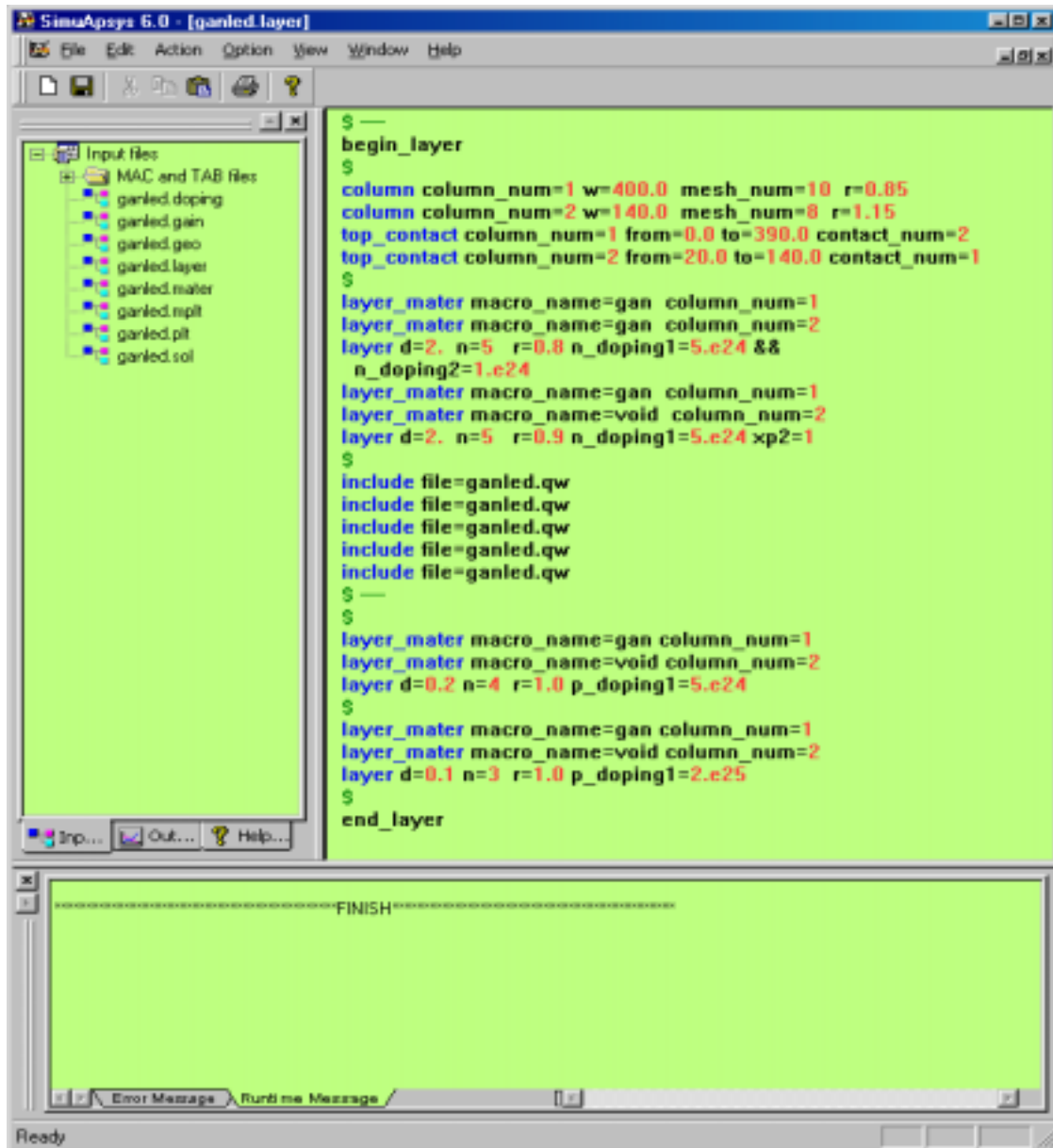


Fig. 13 SimuApsys user interface with ganled.layer

- (3) From the device geometry file (.geo), we now generate the mesh file (.msh). Right-click on “ganled.geo” and select “Generate Mesh”. To view the mesh, right-click on “ganled.mpl”

and select “View Mesh”. You will see a 2D mesh like in Fig. 14. It is very dense in the quantum well region. Close the GSview window.

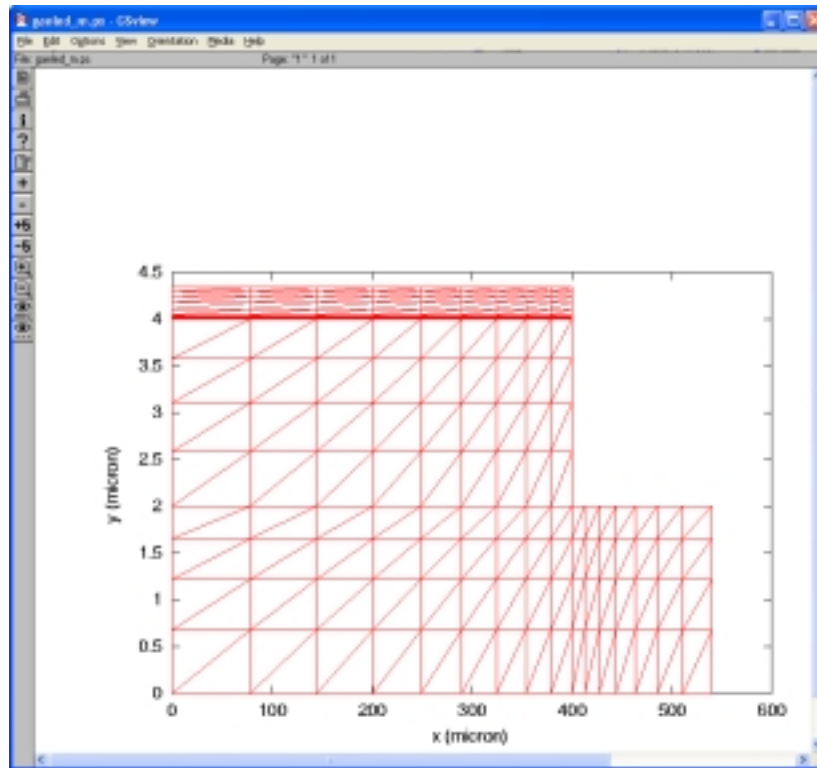


Fig. 14 LED mesh plot.

- (4) In this case, the optical gain is of little interest, however, the spontaneous emission spectrum should be evaluated. Right-click on “ganled.gain” and select “Process and View Gain File”. You will first see the negative gain spectrum (absorption) at low carrier densities. Clicking on “+” in the GSview side panel then shows the spontaneous emission spectrum with a peak at 485nm (Fig. 15). Close the GSview window, both figures are saved as ganled_g.ps
- (5) You can now start the main simulation using the ganled.sol file. Right-click on it and select “Start Simulation”. The simulation ramps up the current according to the settings in ganled.sol. The progress window displays the runtime output which will eventually be saved as ganled.log. It will take only a few minutes. During this time, evaluate the .sol file by double clicking on it. The main command here is “led_control”. No waveguiding effects are considered with “efficiency_model=uniform”. Use the Wizard to check out the command options used. You will find, for instance, that the reflectivity of all surfaces is set to 0.3 which should be modified for a more realistic simulation. Close the wizard after the simulation is completed. Click on “Save” in the pop-up window for the ganled.log file.

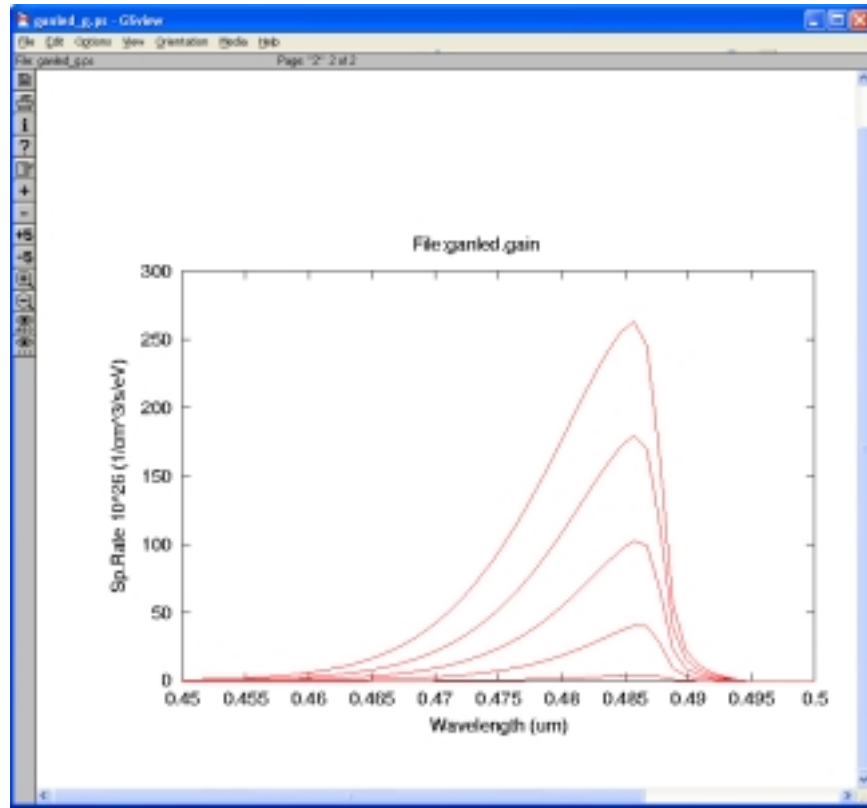


Fig. 15 Spontaneous emission spectrum at different carrier densities.

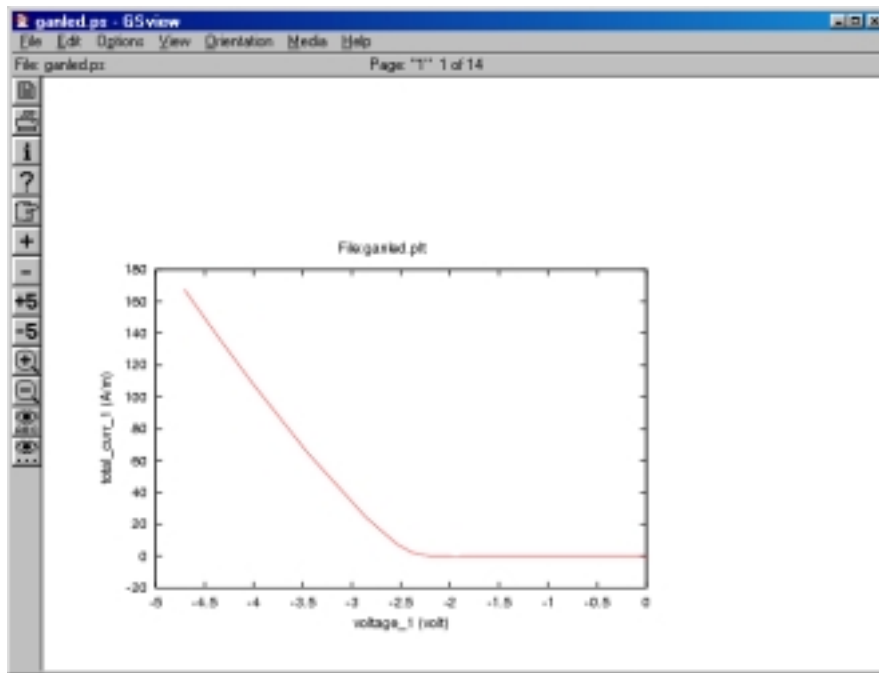


Fig. 16a: LED current-voltage curve.

(6) Double-click on “ganled.plt” to see the plot file. Use the Editing Wizard to check out the plot commands. To plot your results, right-click on “ganled.plt” and select “View Results”. You will see the following plots, some of which indicate strong current crowding near the right edge of column 1 (click on “+” or “-“ to switch):

- current-voltage curve with the current given in A/m for arbitrary device lengths (Fig. 16a)
- total internal light power vs. current with the power given in W/m (Fig. 16b)
- internal efficiency vs. current (Fig. 16c)
- external efficiency for emission from the LED top surface
- external efficiency for emission from the LED bottom surface
- vertical band profile near the left border of the LED with the dense lines indicating the multi-quantum wells
- vertical band profile near the right border of column 1, note difference in potential drop
- 2D surface plot of the electrostatic potential in column 1, the carriers are obviously drawn to the right side
- 2D vector plot of the total current in column 1 with strong current crowding at the right side (Fig. 16d)
- 2D vector plot of the hole current in column 1
- 2D vector plot of the electron current in column 1
- LED emission spectrum
- external efficiency distribution for top emission along lateral direction of column 1 (Fig. 16e)
- far field emission pattern in top direction

Change the plot options or the input files to further explore the device physics. For the same device, self-heating effects are demonstrated in C:\crosslig\apsys_examples\LED\ganled_thermal and ray-tracing is demonstrated in C:\crosslig\apsys_examples\LED_raytrace\dome considering an epoxy dome.

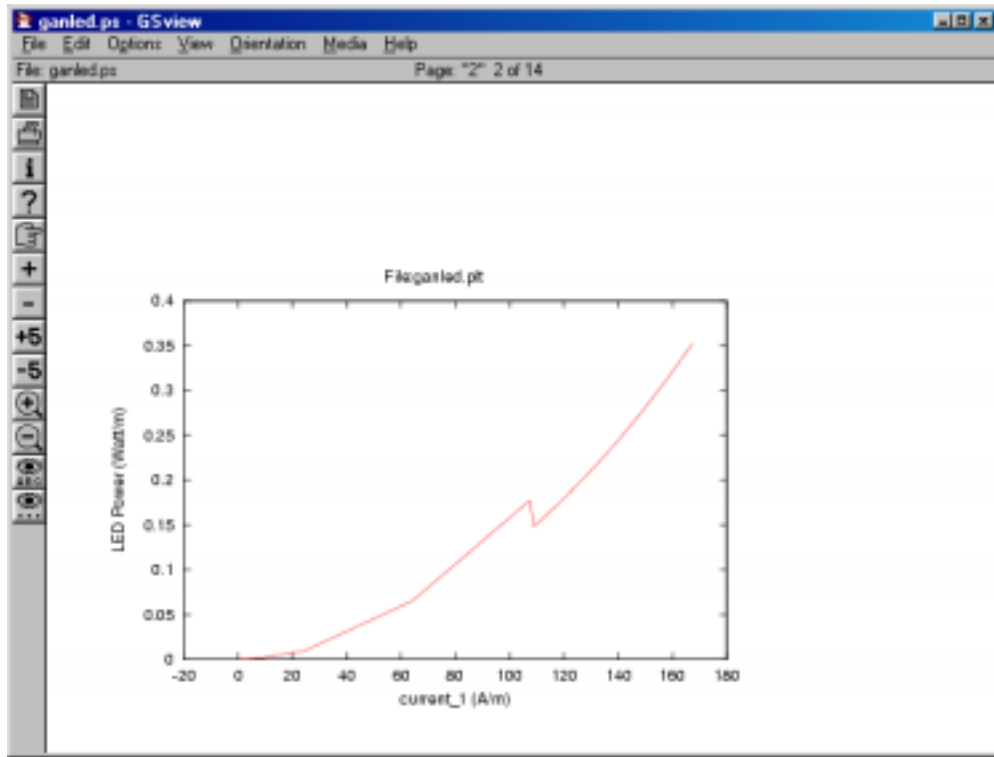


Fig. 16b: Total LED power vs. voltage.

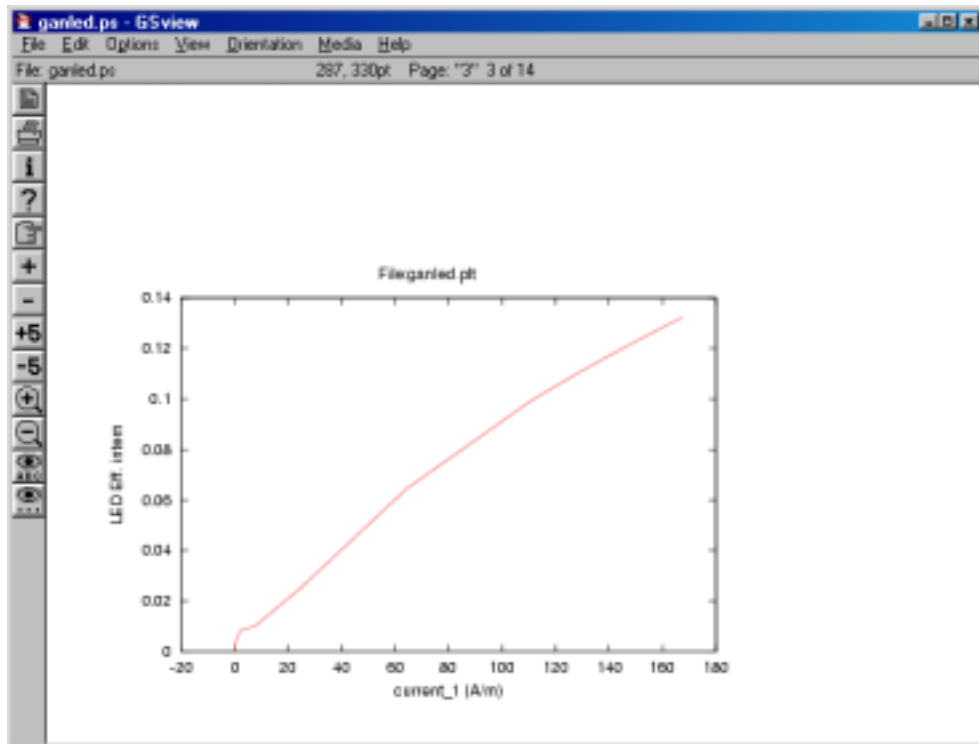


Fig. 16c: Internal LED efficiency vs. current

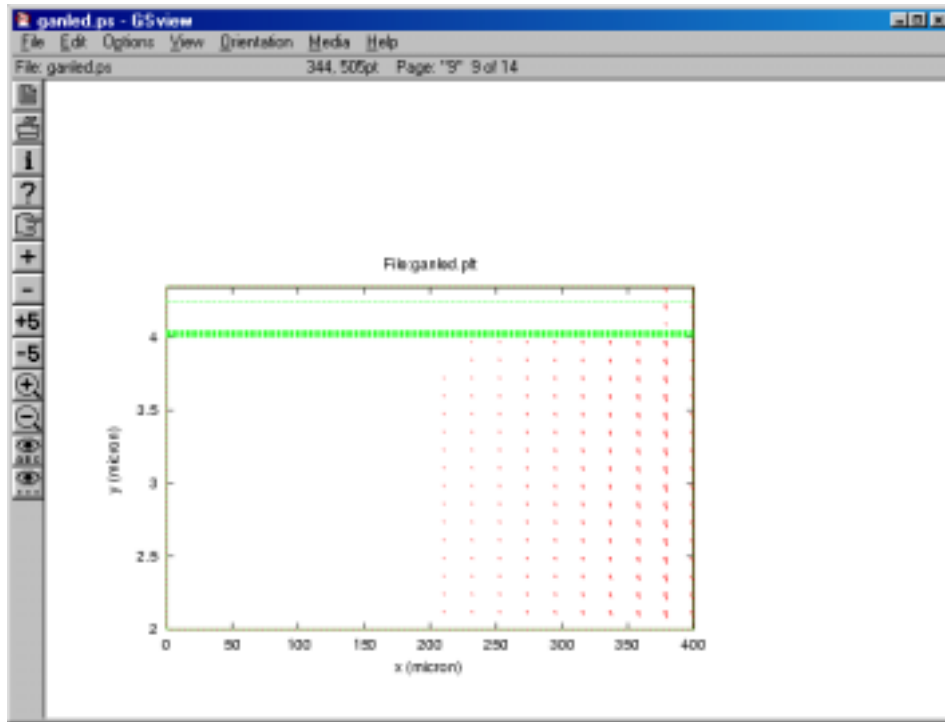


Fig. 16d: Vector plot of current density in column 1 showing strong current crowding on the right.

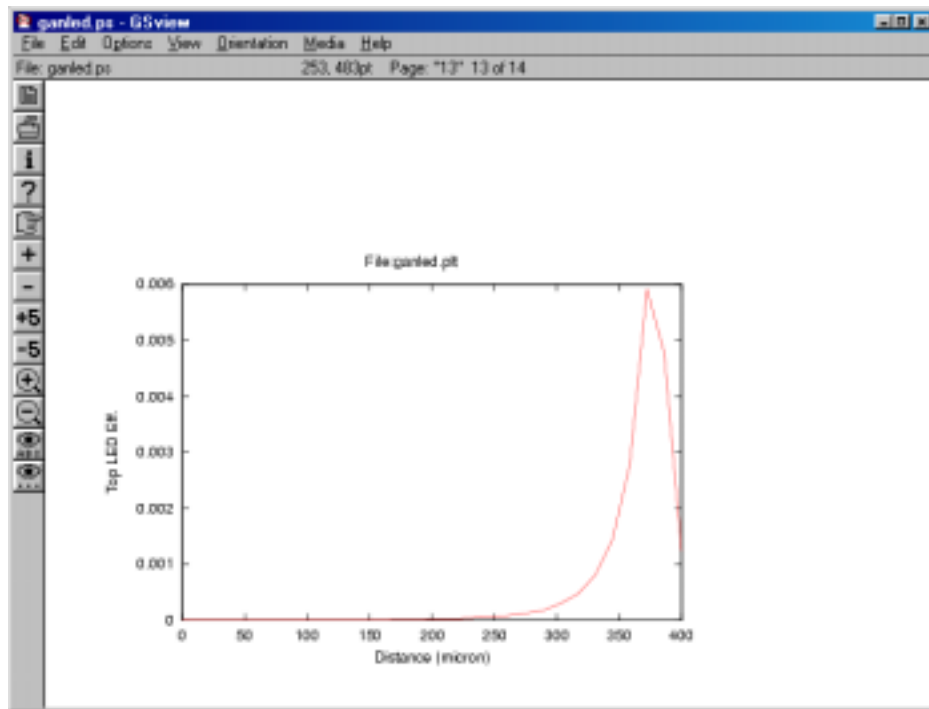


Fig. 16e: External efficiency distribution for top emission along lateral direction of column 1.

5. HOW TO SIMULATE A SIMPLE VERTICAL-CAVITY LASER USING PICS3D

Vertical-cavity surface-emitting lasers (VCSELs) are usually cylinder-symmetric and emit a circular laser beam vertical to the plane of the active pn-junction (Fig. 17). The active region is sandwiched between two distributed Bragg reflectors (DBRs) which form the vertical cavity. DBRs are stacks of two alternating transparent materials with preferably large refractive index contrast to achieve a power reflectivity near 99%.

Compared to in-plane lasers, the simulation of VCSELs differs mainly in the cylindrical coordinate system used and in the calculation of the internal optical field. In longitudinal direction along the VCSEL axis, the optical properties of all layers need to be provided in the *.layer file. This can be done by editing the *.layer file of an in-plane laser or by using “VCSEL SetupLayer” in the SimuPICS3D menu. The latter approach is described in the following. We use a 1.55 μm double-fused VCSEL as example which employs a strained InGaAsP/InP multi-quantum well (MQW) active region and wafer-fused AlGaAs/GaAs DBRs (Fig. 17). The top pillar diameter is 12 μm . For our tutorial simulation, the layer structure is simplified as shown in Tab. 2. In particular, the n-GaAs substrate is neglected and the DBRs are replaced by bulk $\text{Al}_{0.5}\text{Ga}_{0.5}\text{As}$ in the electrical simulation.

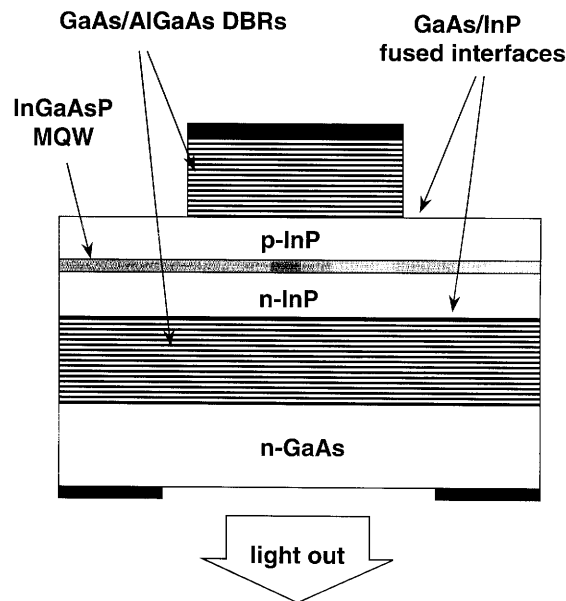


Fig. 17: GaAs/InP double-fused vertical cavity laser

thickness [micron]	doping [m ⁻³]	composition	index	comment
6.9636	p4E23	Al _{0.5} GaAs	2.89/3.38	28x AlAs/GaAs DBR
0.310	p1E24	InP	3.17	p-spacer
0.008	none	InGa _{0.52} As _{0.82} P	3.5	6 quantum barriers
0.0055	none	InGa _{0.24} As _{0.82} P	3.5	7 quantum wells
0.310	n5E24	InP	3.17	n-spacer
7.461	n1E24	Al _{0.5} GaAs	2.89/3.38	30 x AlAs/GaAs DBR

Tab. 2 Simplified layer structure of the vertical-cavity laser (Fig. 17).

This VCSEL is part of the PICS3D example data base (db1.*) in directory C:\crosslig\pics3d_examples\VCSELS\db1_vcsl. In SimuPICS3D, go to “File > Open Project” and load db1.sol. Since this is a 2D problem, the simulation will take significantly more time than in the previous examples. Therefore, we shall setup a 1D version of this VCSEL in the following. Select “File > Close Project”.

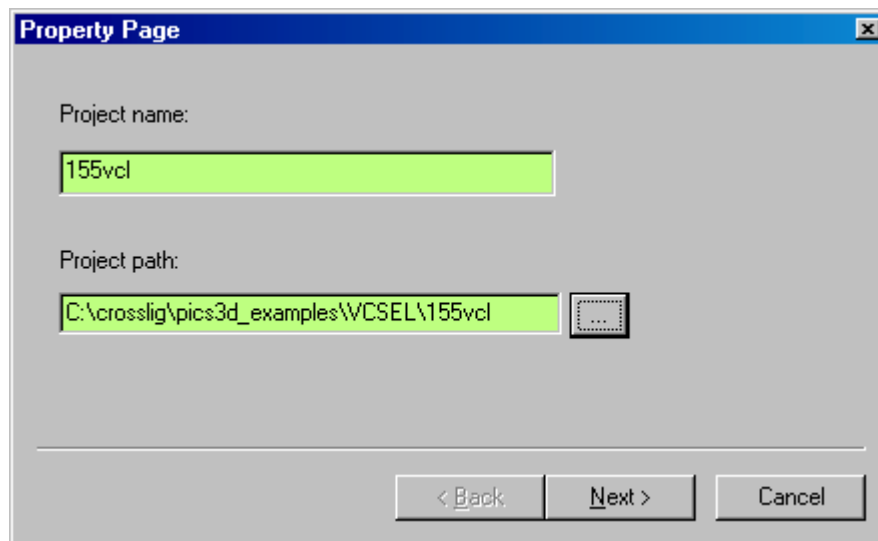


Fig. 17a New project start window..

Select “File > New Project” and enter the information given in Fig. 17a. Click “Next” and select all file types. This will generate all the file names required, however, the files are still empty. From the main menu, select “Action > Setup Device > VCSEL Setup layer”. A window like in

Fig. 17b should appear starting a MS-DOS dialog. Answer all questions carefully using the device information given in this section. Our device has 5 optical VCSEL sections: bottom DBR, n-spacer, MQW, p-spacer, and top DBR. Each optical section can include several layers and the total section thickness is calculated from the sum over all layers in the section. Thus, any thickness input refers to a layer rather than to an optical section. The complete dialog is listed in appendix A5 for your reference.

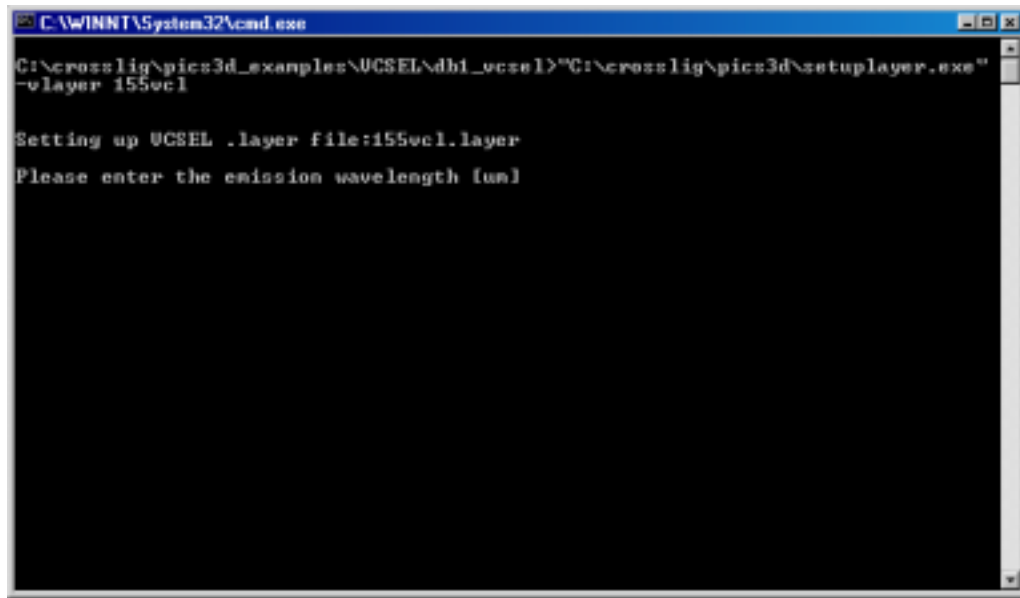


Fig. 17b MS-DOS dialog window for “VCSEL setup layer”

The VCSEL SetupLayer dialog starts with the bottom n-DBR. For the optical model, the software only needs to know the refractive index of both DBR layers and later the total DBR thickness. The thickness and the number of mirror layers is calculated internally. As seen from the bottom, the low index AlAs layer comes first in the bottom DBR (low/high) and the high-index GaAs layer is first in the top DBR (high/low). Use about 10 optical mesh points in each section. In the electrical model, DBRs are treated as an average bulk material, i.e., the average of the layer compositions should be entered.

Two different sets of material macros are required for quantum wells, the “active” macro with bandstructure information etc. and the “bulk” macro with mobility data etc., therefore the composition needs to be entered twice. For the MQW input, use the option “repeat the previous layer” after the first well and the first barrier are introduced. You may repeat these two layers 6 times and later delete the additional barrier layer from the .layer file.

Load and edit your file 155vcl.layer to remove the extra barrier layer and to fix any other typo during your setup. Compared to in-plane lasers, the only new statement is “vcsel_section” which contains optical parameters of each VCSEL section. Please note that the name of each section (“section_type”) must also be used in the corresponding “layer” statements. Check the number n of mesh points in each “layer” statement. The DBRs should have about 20, the spacer layers about 10 and each MQW layer at least 3. Save the file as “155vcl.layer” in your working directory C:\crosslig\pics3d_examples\VCSEL\155vcl.

Processing the layer file produces all the files typical for in-plane lasers plus one new file: 155vcl.vcsel which will be used in the optical part of 155vcl.sol. Check out the mesh. Again, the setup routine produces a 1D structure only, 2D features need to be included by hand. You may zoom into the MQW region by using “yrange= (7.7 7.9)” in the “plot_mesh” command.

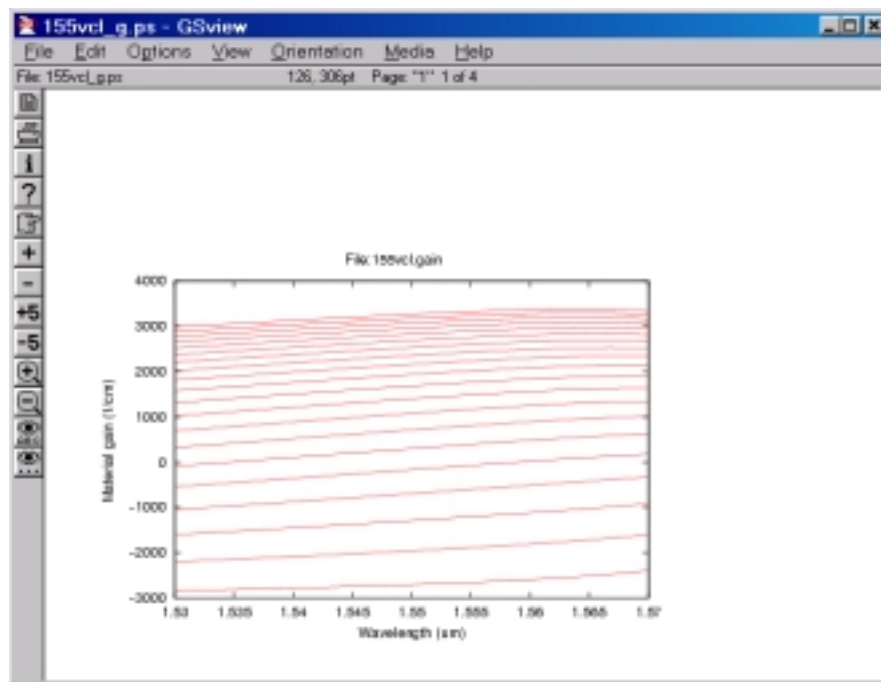


Fig. 18: VCL gain spectrum at different carrier densities.

The remaining files can easily be created using “Action > Main Simulation > Setup .sol file ” in the SimuPICS3D menu. Pick “by command line”. The dialog is given in appendix A6. By default, PICS3D uses Bessel functions for the transversal optical modes which are similar to fiber

modes. The information needed are the core radius ($6\mu\text{m}$), the average core index (3.3), the average cladding index (3.0), as well as the power reflectivity of the bottom AlAs/substrate interface (0.01) and the top AlGaAs/metal interface (0.99). If you have no estimate of the threshold current, just enter a large number (100mA). The background loss is approximated by 3000/m.

After finishing the dialog, process the gain file. The first plot should look similar to Fig. 18. It shows only a small wavelength window near $1.55\mu\text{m}$. The gain becomes quite strong at higher carrier densities, which is important to make this VCL work. In PICS3D, the original gain file should not be manipulated as it is used during the final simulation.

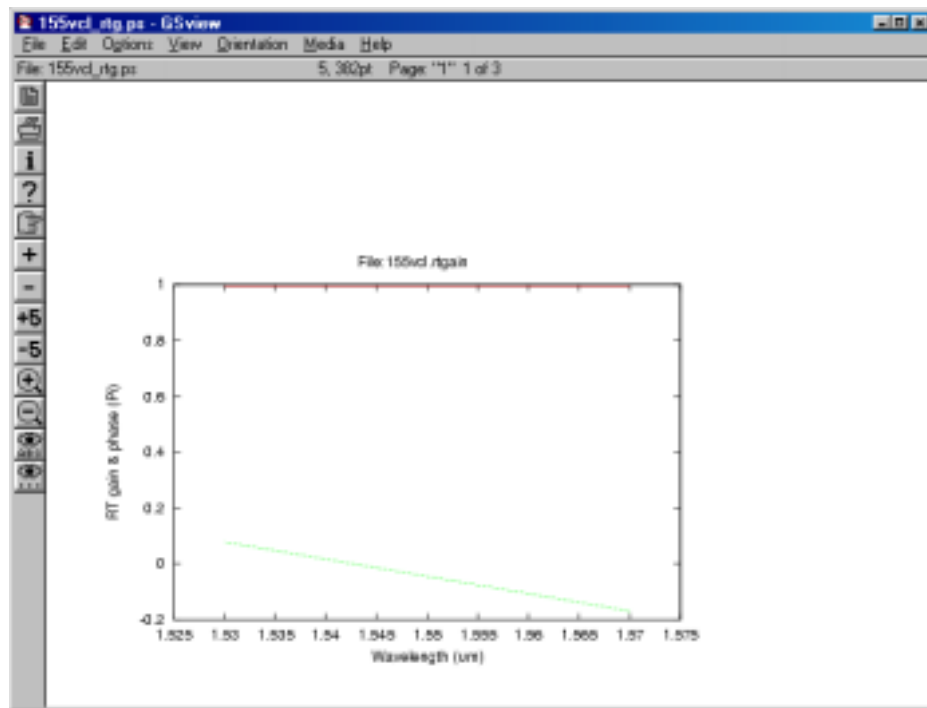


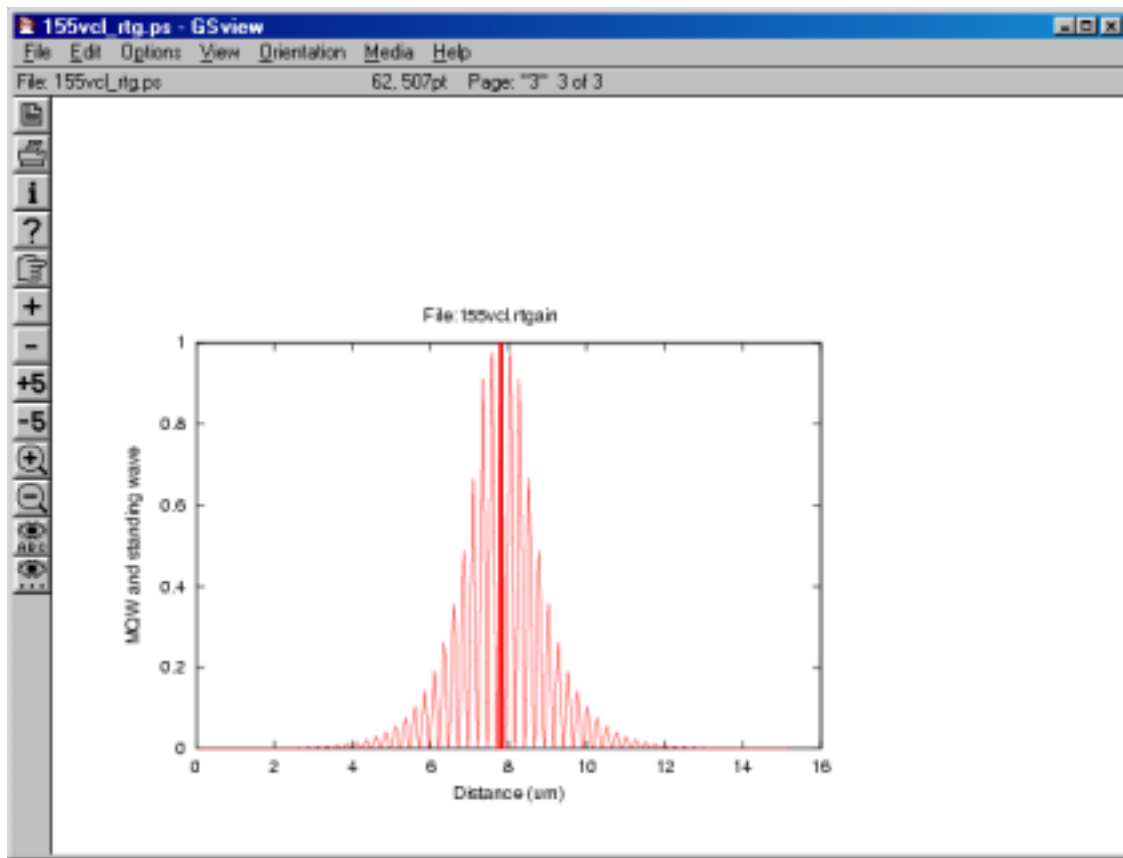
Fig. 19a: Round trip gain and phase (155vcl.rtd)

Like in real life, several VCL design features should be evaluated first before you run the full 3D simulation. Edit and run the 155vcl.sol file as follows:

- 1) Find the emission wavelength: Activate the “stop” in the 155vcl.sol file by removing the \$ sign in front of “stop”. Run the .sol file. It will stop after one minute or so. Save the 155vcl.log file. Right-Click on “155vcl.rtgain” and select “Round Trip Gain”. This plots the 155vcl.rtg file (Fig. 19a). The upper line in Fig. 19a gives the round trip gain and the lower

line gives the round trip phase which is zero for the actual lasing wavelength. In our case, zero phase shift occurs at about 1.54 micron wavelength. This will be the lasing wavelength. If it deviates strongly from the intended wavelength, some mistake occurred in the VCSEL SetupLayer procedure. In this case, the 155vcl.layer file should be edited.

- 2) The third plot of “Round Trip Gain” allows you to check the standing wave overlap with the active region (Fig. 19b). The standing optical wave is plotted on top of the MQW active region. The maximum intensity should be near the center of the MQW, otherwise you need to adjust the spacer layer thickness for better pumping efficiency.



19b: Standing wave and MQW (155vcl.stw).

Deactivate the “stop” in 155vcl.sol by inserting a \$ sign. Start the full simulation, it will take only a few minutes. To check the results, you need to first generate the plot file. Right-click on “155vcl.plt” and select “Setup .plt file”. By default, 10 bias points are used. Select “VCSEL” and the first plot options. Load the plot file and use the Wizard to check out the commands. You may do this while the main simulation is still running.

After the simulation has finished, you can plot your results. Right-click on “155vcl.plt” and select “View Results”. As example, Fig. 20a shows the band diagram, revealing considerable voltage drop in the top DBR at 100mA drive current. Fig. 20b plots the bottom lasing power vs. current, exhibiting a relatively high threshold current of 50 mA, partially caused by the high optical losses. Fig. 20c gives the mode spectrum with a lasing wavelength of about 1.54 μm , as expected from Fig. 19a.

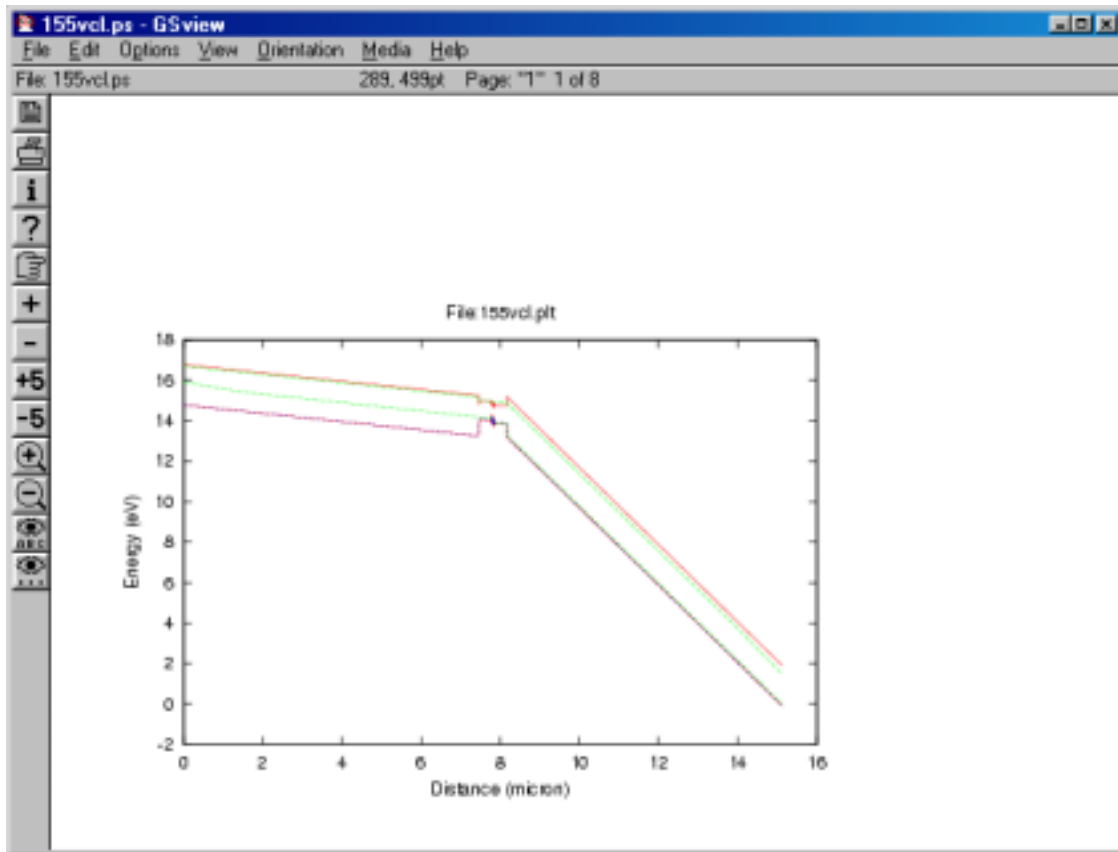


Fig. 20a: VCL band diagram.

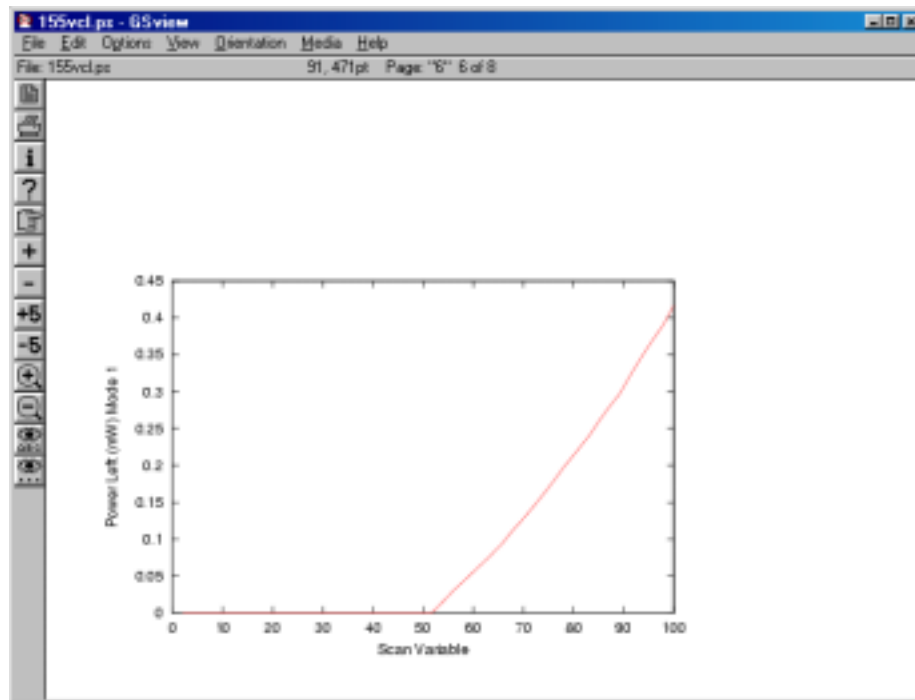


Fig. 20b: Bottom output power vs. current.

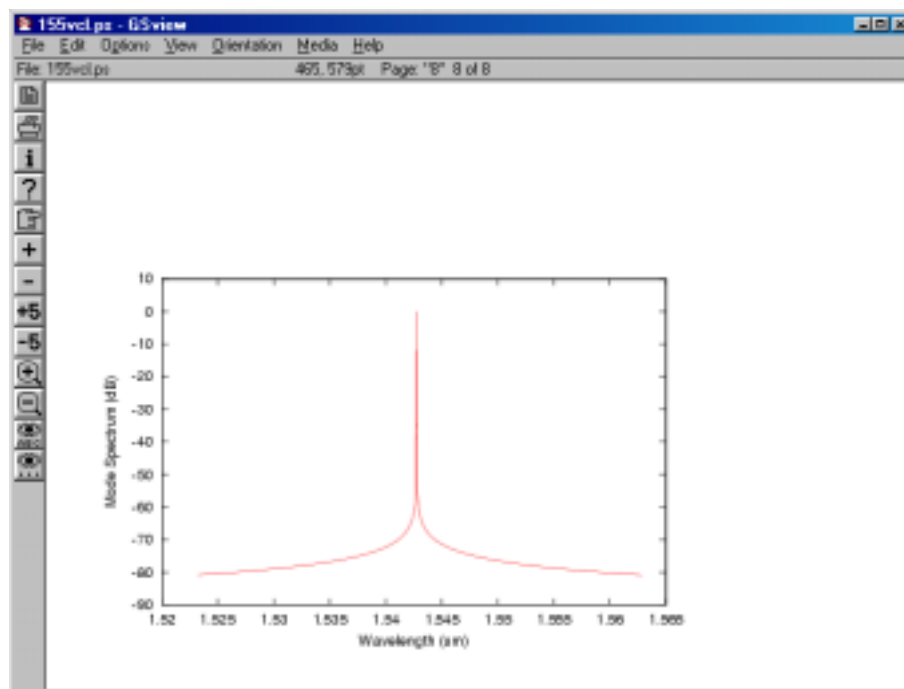


Fig.20c: Mode spectrum.

APPENDIX A1 (“Edge SetupLayer” dialog, see Section 3)

Setting up file:C:\crosslig\lastip_e.layer

We assume a 1D layer structure on the xy-plane.

Please enter the device half-width (um)
 (We assume you have a symmetric device
 and you only need to simulate half of it)
 [typical:1-5 um]

1.5

Please enter the layer structure starting from
 bottom as layer 1

For layer number 1

Please enter layer thickness (um)
 1.35

Please select doping (1/m³):
 n-doping: (1) 2.0e24; (2) 1.5e24
 (3) 1.0e24; (4) 0.5e24; (5) other n-doping
 (6) undoped
 p-doping: (7) 2.0e24; (8) 1.5e24
 (9) 1.0e24; (10) 0.5e24; (11) other p-doping

3

Is this layer an active layer (y/n)?
 n

Please enter layer bulk material macro to be used
 (1) gaas -- bulk GaAs
 (2) inp -- bulk InP
 (3) ingaas -- bulk InGaAs
 (4) algaas -- bulk AlGaAs
 (5) ingaasp -- bulk InGaAsP (matched to InP)
 (6) ingaasp_xy -- bulk InGaAsP (strained)
 (7) Enter another macro name other than the above

4

Do you wish to grade Al [y/n] ?

n

Please enter the Al composition [0-1]
 0.71

Is there grating structure in this layer? [y/n]
 n

Next?

- (1) next layer
- (2) go back to a previous layer
 (so that I can re-enter parameters)
- (3) repeat the previous few layers
 (as in a MQW device)
- (4) generate input files and quit

```

(5) quit
1

For layer number          2

Please enter layer thickness (um)
0.1462

Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
6

Is this layer an active layer (y/n)?
n

Please enter layer bulk material macro to be used
(1) gaas -- bulk GaAs
(2) inp  -- bulk InP
(3) ingaas -- bulk InGaAs
(4) algaas -- bulk AlGaAs
(5) ingaasp -- bulk InGaAsP (matched to InP)
(6) ingaasp_xy -- bulk InGaAsP (strained)
(7) Enter another macro name other than the above
4

Do you wish to grade Al [y/n] ?
y
Please enter the starting Al composition [0-1]
0.71
Please enter the ending Al composition [0-1]
0.33

Is there grating structure in this layer? [y/n]
n
Next?
(1) next layer
(2) go back to a previous layer
    (so that I can re-enter parameters)
(3) repeat the previous few layers
    (as in a MQW device)
(4) generate input files and quit
(5) quit
1

For layer number          3

Please enter layer thickness (um)
0.0076

Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24

```

```

        (9) 1.0e24; (10) 0.5e24; (11) other p-doping
6
Is this layer an active layer (y/n)?
y
Please enter active layer material macro
quantum subbands, gain, etc.

Please enter the active region material system
(1) AlGaAs/AlGaAs --- quantum well
(2) InGaAs/AlGaAs --- strained quantum well
(3) InGaAsP/InP --- InGaAsP/InGaAsP strained
    quantum well/barrier; substrate=InP.
(4) AlGaAs --- bulk AlGaAs
(5) InGaAsP --- bulk InGaAsP lattice matched to InP
(6) InGaAsP/GaAs --- InGaAsP/InGaAsP strained
    quantum well/barrier; substrate=GaAs.
(7) InGaAlAs/InP --- InGaAlAs/InGaAlAs strained
    quantum well/barrier; substrate=InP.
(8) GaN --- bulk GaN (9) AlGaN --- bulk AlGaN
(10) InGaN --- bulk InGaN
(11) InGaN/AlGaN -- strained well/barrier;
    substrate = GaN.
(12) InGaAlP/GaAs --- In(1-xw-yw)Ga(xw)Al(yw)P
    /In(1-xb-yb)Ga(xb)Al(yb)P; substrate=GaAs
(13) InGaN/InGaN --- Strained well grown on material
    lattice matched to GaN.
1
Enter Al composition in well (0 to 1)
0
Enter Al composition in barrier (0 to 1)
0.33
Enter Temperature (Kelvin)
300

Please enter layer bulk material macro to be used
(1) gaas -- bulk GaAs
(2) inp -- bulk InP
(3) ingaas -- bulk InGaAs
(4) algaas -- bulk AlGaAs
(5) ingaasp -- bulk InGaAsP (matched to InP)
(6) ingaasp_xy -- bulk InGaAsP (strained)
(7) Enter another macro name other than the above
1

Is there grating structure in this layer? [y/n]
n
Next?
(1) next layer
(2) go back to a previous layer
    (so that I can re-enter parameters)
(3) repeat the previous few layers
    (as in a MQW device)
(4) generate input files and quit
(5) quit
1

```

```

For layer number          4

Please enter layer thickness (um)
0.1462

Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
6

Is this layer an active layer (y/n)?
n

Please enter layer bulk material macro to be used
(1) gaas -- bulk GaAs
(2) inp -- bulk InP
(3) ingaas -- bulk InGaAs
(4) algaas -- bulk AlGaAs
(5) ingaasp -- bulk InGaAsP (matched to InP)
(6) ingaasp_xy -- bulk InGaAsP (strained)
(7) Enter another macro name other than the above
4
Do you wish to grade Al [y/n] ?
y
Please enter the starting Al composition [0-1]
0.33
Please enter the ending Al composition [0-1]
0.71

Is there grating structure in this layer? [y/n]
n
Next?
(1) next layer
(2) go back to a previous layer
    (so that I can re-enter parameters)
(3) repeat the previous few layers
    (as in a MQW device)
(4) generate input files and quit
(5) quit
1

```

```

For layer number          5

Please enter layer thickness (um)
1.35

Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
9

```

```

Is this layer an active layer (y/n)?
n

Please enter layer bulk material macro to be used
(1) gaas -- bulk GaAs
(2) inp -- bulk InP
(3) ingaas -- bulk InGaAs
(4) algaas -- bulk AlGaAs
(5) ingaasp -- bulk InGaAsP (matched to InP)
(6) ingaasp_xy -- bulk InGaAsP (strained)
(7) Enter another macro name other than the above
4
Do you wish to grade Al [y/n] ?
n
Please enter the Al composition [0-1]
0.71

Is there grating structure in this layer? [y/n]
n
Next?
(1) next layer
(2) go back to a previous layer
    (so that I can re-enter parameters)
(3) repeat the previous few layers
    (as in a MQW device)
(4) generate input files and quit
(5) quit
4
Please enter the total number of mesh points
    for all the layers; typical [20-40]
20
Enter the maximum average mesh spacing:
    typical [0.1 - 0.5 um]
.2

Generating .layer file

```

APPENDIX A2 (Setup .gain dialog, see Section 3)

```

Setting up input file: mylaser.gain
You may use it to plot the gain/loss spectrum
    and other physical properties

Enter estimated operating wavelength in um
    for example 0.82:
0.83
Enter carrier density range in m**(-3)
    for example:
    5.e23 1.e25
5e23 1e25
Enter temperature in degrees Kelvin

```



```

300
  Please enter the graphics device
    (1) x11 (2) postscript (3) vttek (4) window
  Your choice?
2
  Please select the physical properties to plot
    (1) gain spectrum
    (2) spontaneous emission rate
    (3) index change spectrum
    (4) current vs. active region density
    (5) alpha factor spectrum
    (6) gain vs. active region density
    (7) quantum well subbands (k.p)
    (8) dipole moment vs. k (k.p)
    (9) 1 - 4 above
    (10) 1 - 6 above
    (11) finish
1
  Enter your next choice (1-11)
2
  Enter your next choice (1-11)
6
  Enter your next choice (1-11)
11

Generating .gain file

```

APPENDIX A3 (Setup .sol dialog, see Section 3)

```

Setting up .sol file for filebase=mylaser
We assume that you already generated
  .geo, .mater and .doping

Setting wave equation boundary:
  Enter 1 to solve wave in the whole device
  Enter 2 to set boundary manually
1
  Please enter the cavity length (um):
400
  Please enter the expected emission wavelength (um):
0.83
  Please enter 1 for broad-area (1D) laser wave boundary
  Please enter 2 for other 2D laser boundary
1
  Please enter the current bias you wish to apply (mA)
10
  Please enter the contact number for the n-type laser contact
1

Generating .sol file

```

APPENDIX A4 (Setup .plt file dialog, see Section 3)

Setting up .plt file for filebase=mylaser

Please enter the data sets you wish to plot
[Each data set corresponds to a bias condition]
data_set?

3

Please enter the maximum data sets available
[so that bias dependent plot can be made]
max_data_set?

3

Please enter the graphics device
(1) x11 (2) postscript (3) vttek (4) window
Your choice?

2

Generating .plt file

Adding a new plot to .plt file
Please select the type of plot:

- (1) 1d plot along y
- (2) 1d plot along x
- (3) 2d contour plot
- (4) 2d vector plot
- (5) 3d surface plot
- (6) I-V characteristics
- (7) I-t characteristics
- (8) L-I characteristics
- (9) quit

plot_type?

1

Please enter the x-position of the 1d plot

- (1) at 0
- (2) at 1/10 xsize
- (3) at 1/4 xsize
- (4) at 1/2 xsize
- (5) at 3/4 xsize
- (6) at xsize

x-position?

4

Please enter the variable you wish to plot

- (1) electron concentration (2) hole concentration
- (3) trap concentration (4) potential
- (5) both electron & hole conc. (6) energy bands
- (7) elec current [x-component] (8) elec current [y-component]
- (9) hole current [x-comp] (10) hole current [y-comp]
- (11) displac. curr. [x-comp] (12) displac. curr. [y-comp]
- (13) total current [x-comp] (14) total current [y-comp]
- (15) x-comp. electric field (16) y-comp. electric field
- (17) trap occupancy (18) optical generation rate
- (19) optical field (20) hot electron energy
- (21) donor concentration (22) acceptor concentration
- (23) radiative recombination (24) Auger recombination
- (25) SRH recombination (26) lattice temperature

(27) electron conc difference between nodes
 (28) wave intensity (29) local gain
 (30) stimulated recombination
 variable?

6

Adding a new plot to .plt file
 Please select the type of plot:

- (1) 1d plot along y
- (2) 1d plot along x
- (3) 2d contour plot
- (4) 2d vector plot
- (5) 3d surface plot
- (6) I-V characteristics
- (7) I-t characteristics
- (8) L-I characteristics
- (9) quit

plot_type?

1

Please enter the x-position of the 1d plot

- (1) at 0
- (2) at 1/10 xsize
- (3) at 1/4 xsize
- (4) at 1/2 xsize
- (5) at 3/4 xsize
- (6) at xsize

x-position?

4

Please enter the variable you wish to plot

- (1) electron concentration (2) hole concentration
- (3) trap concentration (4) potential
- (5) both electron & hole conc. (6) energy bands
- (7) elec current [x-component] (8) elec current [y-component]
- (9) hole current [x-comp] (10) hole current [y-comp]
- (11) displac. curr. [x-comp] (12) displac. curr. [y-comp]
- (13) total current [x-comp] (14) total current [y-comp]
- (15) x-comp. electric field (16) y-comp. electric field
- (17) trap occupancy (18) optical generation rate
- (19) optical field (20) hot electron energy
- (21) donor concentration (22) acceptor concentration
- (23) radiative recombination (24) Auger recombination
- (25) SRH recombination (26) lattice temperature
- (27) electron conc difference between nodes
- (28) wave intensity (29) local gain
- (30) stimulated recombination

variable?

29

Adding a new plot to .plt file
 Please select the type of plot:

- (1) 1d plot along y
- (2) 1d plot along x
- (3) 2d contour plot
- (4) 2d vector plot
- (5) 3d surface plot
- (6) I-V characteristics

```

      (7) I-t characteristics
      (8) L-I characteristics
      (9) quit
plot_type?
8

Please enter the electrode number for I
I_electrode_num?
1

Adding a new plot to .plt file
Please select the type of plot:
  (1) 1d plot along y
  (2) 1d plot along x
  (3) 2d contour plot
  (4) 2d vector plot
  (5) 3d surface plot
  (6) I-V characteristics
  (7) I-t characteristics
  (8) L-I characteristics
  (9) quit
plot_type?
9

Generating .plt file

```

APPENDIX A5 (“VCSEL SetupLayer” dialog, see Section 5)

```

Please enter the emission wavelength [um]
1.55
Please enter the two indices
  of the DBR mirror; example [3.6 3.3]
2.89 3.38
We assume a 1-column layer structure.

Please enter the device half-width (um)
(We assume you have a symmetric device
  and you only need to simulate half of it)
[typical:1-5 um]
6

```

```

Start a new VCSEL section which may include one
or more material layers.
Please enter the optical grating model:
  (1) non-mirror section
  (2) DBR mirror with high/low indices
  (3) DBR mirror with low/high indices
3
Please enter the optical mesh number
  for this VCSEL section. Typical [4-20]

```

10

For layer number 1

Please enter layer thickness (um)

6.972

Please select doping (1/m³):

n-doping: (1) 2.0e24; (2) 1.5e24

(3) 1.0e24; (4) 0.5e24; (5) other n-doping

(6) undoped

p-doping: (7) 2.0e24; (8) 1.5e24

(9) 1.0e24; (10) 0.5e24; (11) other p-doping

3

Please enter layer bulk material macro to be used

(1) gaas -- bulk GaAs

(2) inp -- bulk InP

(3) ingaas -- bulk InGaAs

(4) algaas -- bulk AlGaAs

(5) ingaasp -- bulk InGaAsP (matched to InP)

(6) ingaasp_xy -- bulk InGaAsP (strained)

(7) Enter another macro name other than the above

4

Please enter the Al composition [0-1]

.5

Next?

(1) next layer

(2) go back to a previous layer
(so that I can re-enter parameters)(3) repeat the previous few layers
(as in a MQW device)

(4) generate input files and quit

(5) quit

1

Shall we start a new VCSEL section [y/n]?

y

Start a new VCSEL section which may include one
or more material layers.

Please enter the optical grating model:

(1) non-mirror section

(2) DBR mirror with high/low indices

(3) DBR mirror with low/high indices

1

Please enter the optical mesh number
for this VCSEL section. Typical [4-20]

10

Does this section contain active layer(s) [y/n]?

n

For layer number 2

```

Please enter layer thickness (um)
.31

Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
4
Please enter layer bulk material macro to be used
(1) gaas -- bulk GaAs
(2) inp  -- bulk InP
(3) ingaas -- bulk InGaAs
(4) algaas -- bulk AlGaAs
(5) ingaasp -- bulk InGaAsP (matched to InP)
(6) ingaasp_xy -- bulk InGaAsP (strained)
(7) Enter another macro name other than the above
2
Next?
(1) next layer
(2) go back to a previous layer
    (so that I can re-enter parameters)
(3) repeat the previous few layers
    (as in a MQW device)
(4) generate input files and quit
(5) quit
1
Shall we start a new VCSEL section [y/n]?
y
Start a new VCSEL section which may include one
or more material layers.
Please enter the optical grating model:
(1) non-mirror section
(2) DBR mirror with high/low indices
(3) DBR mirror with low/high indices
1
Please enter the optical mesh number
for this VCSEL section. Typical [4-20]
10
Does this section contain active layer(s) [y/n]?
y

For layer number          3

Please enter layer thickness (um)
.0055

Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping

```

```

        (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24
          (9) 1.0e24; (10) 0.5e24; (11) other p-doping

```

```

6
Is this layer an active layer (y/n)?
y

```

```

Please enter active layer material macro
for quantum subbands, gain, etc.

```

```

Please enter the active region material system

```

```

(1) AlGaAs/AlGaAs --- quantum well
(2) InGaAs/AlGaAs --- strained quantum well
(3) InGaAsP/InP --- InGaAsP/InGaAsP strained
    quantum well/barrier; substrate=InP.
(4) AlGaAs --- bulk AlGaAs
(5) InGaAsP --- bulk InGaAsP lattice matched to InP
(6) InGaAsP/GaAs --- InGaAsP/InGaAsP strained
    quantum well/barrier; substrate=GaAs.
(7) InGaAlAs/InP --- InGaAlAs/InGaAlAs strained
    quantum well/barrier; substrate=InP.
(8) GaN --- bulk GaN (9) AlGaN --- bulk AlGaN
(10) InGaN --- bulk InGaN
(11) InGaN/AlGaN -- strained well/barrier;
    substrate = GaN.
(12) InGaAlP/GaAs --- In(1-xw-yw)Ga(xw)Al(yw)P
    /In(1-xb-yb)Ga(xb)Al(yb)P; substrate=GaAs
(13) InGaN/InGaN --- Strained well grown on material
    lattice matched to GaN.

```

```

3
Enter Ga composition in well (0 to 1)
.24
Enter As composition in well (0 to 1)
.82
Enter Ga composition in barrier (0 to 1)
.52
Enter As composition in barrier (0 to 1)
.82
Enter Temperature (Kelvin)
300

```

```

Please enter layer bulk material macro to be used

```

```

(1) gaas -- bulk GaAs
(2) inp -- bulk InP
(3) ingaas -- bulk InGaAs
(4) algaas -- bulk AlGaAs
(5) ingaasp -- bulk InGaAsP (matched to InP)
(6) ingaasp_xy -- bulk InGaAsP (strained)
(7) Enter another macro name other than the above

```

```

6
Please enter the Ga and As composition [0-1, 0-1]
.24 .82

```

```

Next?
  (1) next layer
  (2) go back to a previous layer
      (so that I can re-enter parameters)
  (3) repeat the previous few layers
      (as in a MQW device)
  (4) generate input files and quit
  (5) quit
1
  Shall we start a new VCSEL section [y/n]?
n

  For layer number          4

  Please enter layer thickness (um)
.008

  Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
6
  Is this layer an active layer (y/n)?
n
  Please enter layer bulk material macro to be used
  (1) gaas -- bulk GaAs
  (2) inp  -- bulk InP
  (3) ingaas -- bulk InGaAs
  (4) algaas -- bulk AlGaAs
  (5) ingaasp -- bulk InGaAsP (matched to InP)
  (6) ingaasp_xy -- bulk InGaAsP (strained)
  (7) Enter another macro name other than the above
6
  Please enter the Ga and As composition [0-1, 0-1]
.52 .82
Next?
  (1) next layer
  (2) go back to a previous layer
      (so that I can re-enter parameters)
  (3) repeat the previous few layers
      (as in a MQW device)
  (4) generate input files and quit
  (5) quit
3
  How many number of previous layers to repeat?
  (must be > 1)
2
  For how many times?
6

```



```

Next?
  (1) next layer
  (2) go back to a previous layer
      (so that I can re-enter parameters)
  (3) repeat the previous few layers
      (as in a MQW device)
  (4) generate input files and quit
  (5) quit
1
  Shall we start a new VCSEL section [y/n]?
y
  Start a new VCSEL section which may include one
  or more material layers.
  Please enter the optical grating model:
    (1) non-mirror section
    (2) DBR mirror with high/low indices
    (3) DBR mirror with low/high indices
1
  Please enter the optical mesh number
  for this VCSEL section. Typical [4-20]
10
  Does this section contain active layer(s) [y/n]?
n

  For layer number          17

  Please enter layer thickness (um)
.31

  Please select doping (1/m^3):
n-doping: (1) 2.0e24; (2) 1.5e24
           (3) 1.0e24; (4) 0.5e24; (5) other n-doping
           (6) undoped
p-doping: (7) 2.0e24; (8) 1.5e24
           (9) 1.0e24; (10) 0.5e24; (11) other p-doping
9
  Please enter layer bulk material macro to be used
  (1) gaas -- bulk GaAs
  (2) inp -- bulk InP
  (3) ingaas -- bulk InGaAs
  (4) algaas -- bulk AlGaAs
  (5) ingaasp -- bulk InGaAsP (matched to InP)
  (6) ingaasp_xy -- bulk InGaAsP (strained)
  (7) Enter another macro name other than the above
2
Next?
  (1) next layer
  (2) go back to a previous layer
      (so that I can re-enter parameters)
  (3) repeat the previous few layers
      (as in a MQW device)

```

```

    (4) generate input files and quit
    (5) quit
1
  Shall we start a new VCSEL section [y/n]?
y
  Start a new VCSEL section which may include one
  or more material layers.
  Please enter the optical grating model:
    (1) non-mirror section
    (2) DBR mirror with high/low indices
    (3) DBR mirror with low/high indices
2
  Please enter the optical mesh number
  for this VCSEL section. Typical [4-20]
10

  For layer number          18

  Please enter layer thickness (um)
6.507

  Please select doping (1/m^3):
  n-doping: (1) 2.0e24; (2) 1.5e24
            (3) 1.0e24; (4) 0.5e24; (5) other n-doping
            (6) undoped
  p-doping: (7) 2.0e24; (8) 1.5e24
            (9) 1.0e24; (10) 0.5e24; (11) other p-doping
11
  Please enter the p-doping concentration
4e23
  Please enter layer bulk material macro to be used
    (1) gaas -- bulk GaAs
    (2) inp -- bulk InP
    (3) ingaas -- bulk InGaAs
    (4) algaas -- bulk AlGaAs
    (5) ingaasp -- bulk InGaAsP (matched to InP)
    (6) ingaasp_xy -- bulk InGaAsP (strained)
    (7) Enter another macro name other than the above
4
  Please enter the Al composition [0-1]
.5
Next?
    (1) next layer
    (2) go back to a previous layer
        (so that I can re-enter parameters)
    (3) repeat the previous few layers
        (as in a MQW device)
    (4) generate input files and quit
    (5) quit
4
  Please enter the total number of mesh points

```

```

    for all the layers; typical [20-40]
40
    Enter the maximum average mesh spacing:
    typical [0.1 - 0.5 um]
1.0

    Generating .layer

```

APPENDIX A6 (Setup .sol dialog for VCSEL, see Section 5)

```

    Setting up the software files with filebase=155vcl
    Is the laser type (1) edge or (2) VCSEL?
2
    Please enter the operating wavelength (um)
    (it should also be the Bragg wavelength, if applicable)
1.55

    Using the effective index method, we model the
    lateral mode by a co-axial waveguide model.
    Please enter the core radius
6
    Please enter the core index
3.3
    Please enter the cladding index (outside the core)
3.0
    Please enter the bottom and top
    facet power reflectivities (2 numbers)
    Typical [0.3 0.3]
0.01 0.99

    Please enter n-contact number
    (1) n-contact at bottom (2) on top
1
    Please enter your estimate of
    threshold current (mA)
100
    Please enter the background loss [1/m]
3000

    Generate .sol file
    Generate .gain file
    Generate .mplt file
    Generate .rtg file

```

THE END